

Taxation of the Factor Capital

Economic Impact Analysis of Government Measures



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- **Overview**
- Introduction
- Intertemporal Consumption Decisions (intertemporal consumption)
 - Taxation in the Two-Period Model (Income Tax, Capital Gains Tax, Wealth Tax)
- Optimal Taxation of the Factor Capital
 - Theory
 - Empirical Evidence
 - Distributional Aspects
 - Political Implications
 - Limitations and Extensions (Behavioral Economics)

Taxation of the Factor Capital - Introduction



Saez – Course Public Economics (<https://eml.berkeley.edu/~saez/course131/course131.html>)

- **How rich is one with**
 - 200.000 USD,
 - 1 Mio USD,
 - 50 Mio USD und
 - 1 Mrd USD **in the USA?**

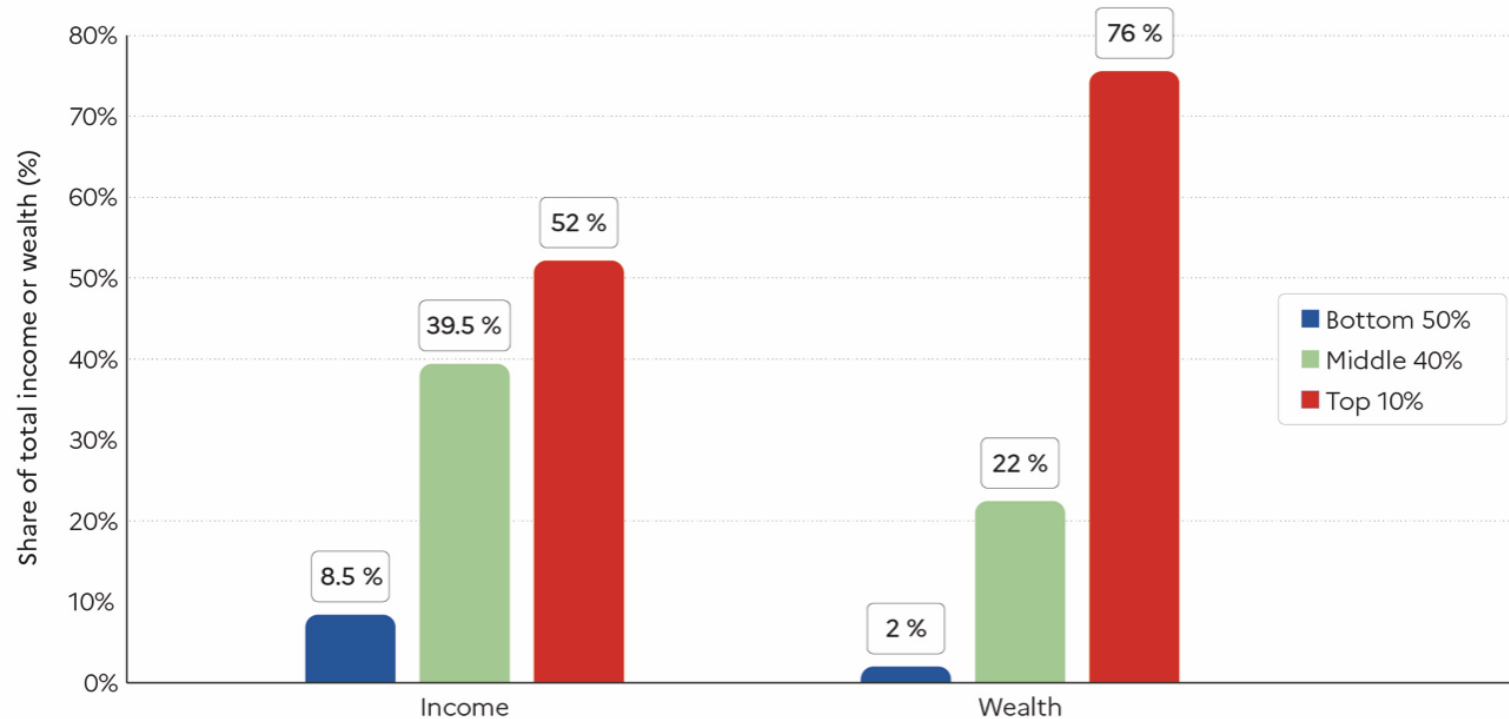


	Threshold	Percent_Above	People_Above
\$200K	\$200K	49.18516201	103,288,840
\$1M	\$1M	18.03795335	37,879,702
\$50M	\$50M	0.14089450	295,878
\$1B	\$1B	0.00126024	2,647

- What percentage of **income** does the **top 10%** have? (Global)
- What percentage of **capital** (wealth) does the **top 10%** have? (Global)

Income and Capital Distribution

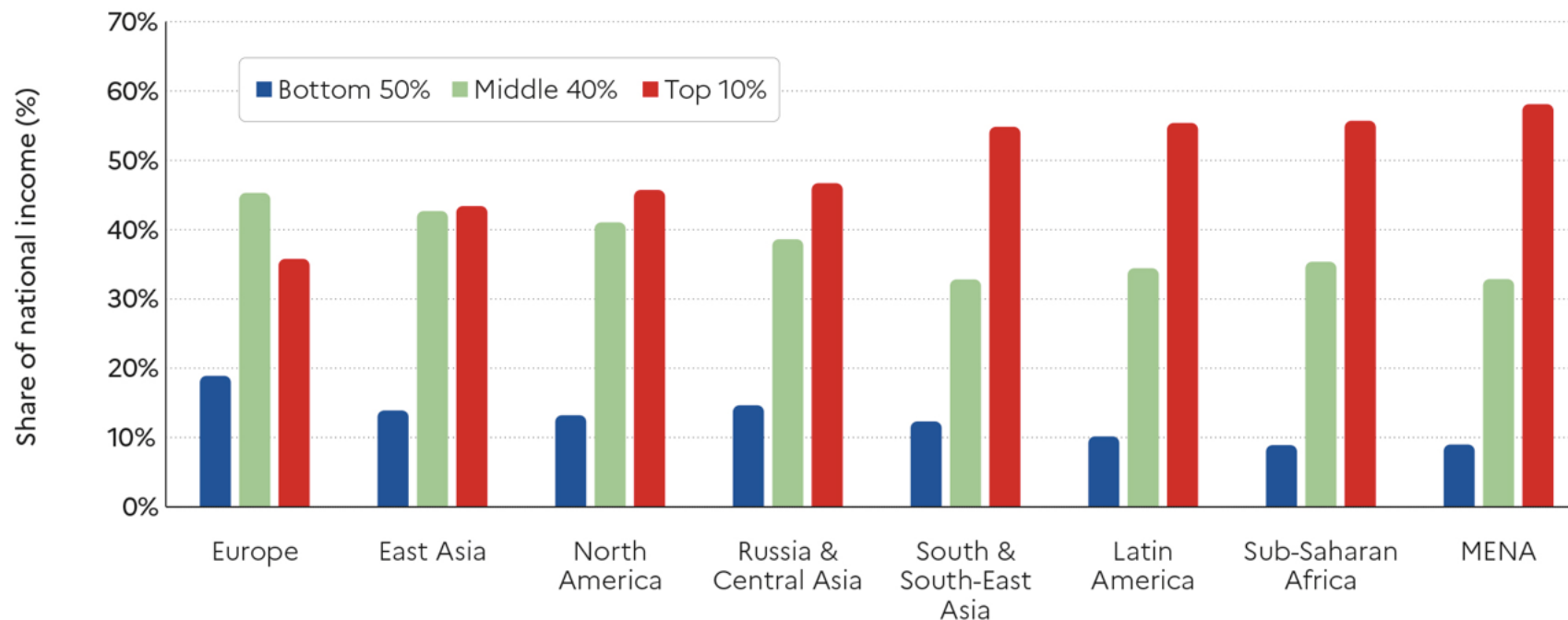
Figure 1 Global income and wealth inequality, 2021



Interpretation: The global bottom 50% captures 8.5% of total income measured at Purchasing Power Parity (PPP). The global bottom 50% owns 2% of wealth (at Purchasing Power Parity). The global top 10% owns 76% of total Household wealth and captures 52% of total income in 2021. Note that top wealth holders are not necessarily top income holders. Incomes are measured after the operation of pension and unemployment systems and before taxes and transfers. **Sources and series:** wii2022.wid.world/methodology.

Income Distribution Very Different

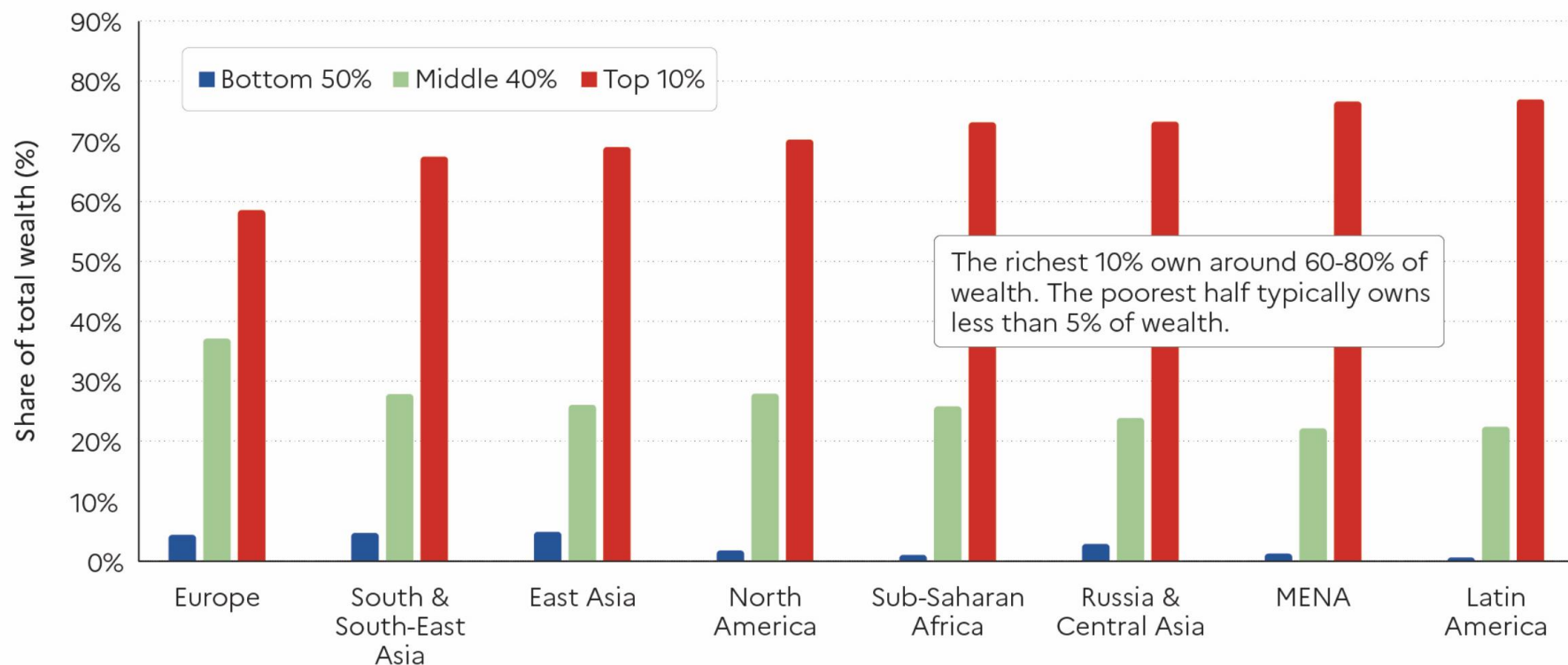
Figure 2 The poorest half lags behind: Bottom 50%, middle 40% and top 10% income shares across the world in 2021



Interpretation: In Latin America, the top 10% captures 55% of national income, compared to 36% in Europe. Income is measured after pension and unemployment contributions and benefits paid and received by individuals but before income taxes and other transfers. **Sources and series:** www.wir2022.wid.world/methodology.

Capital Distribution Even More Uneven

Figure 4 The extreme concentration of capital: wealth inequality across the world, 2021

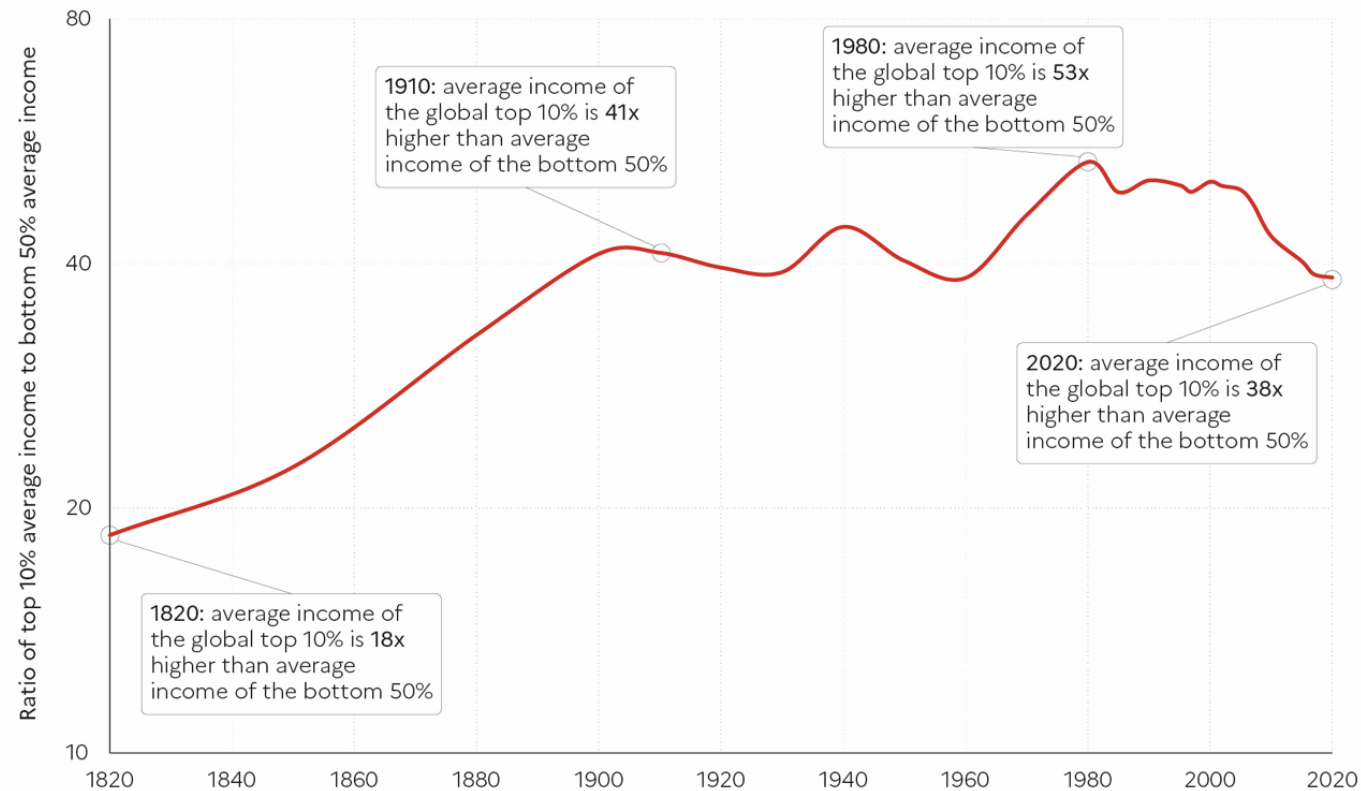


Interpretation: The top 10% in Latin America captures 77% of total household wealth, compared with 1% captured by the bottom 50%. Net household wealth is equal to the sum of financial assets (e.g. equity or bonds) and non-financial assets (e.g. housing or land) owned by individuals, net of their debts. **Sources and series:** wir2022.wid.world/methodology.

- How has **Global Inequality (P10/P50)** developed over the **last 20 years**?

Declining Global Inequality

Figure 5 Global income inequality: T10/B50 ratio, 1820-2020

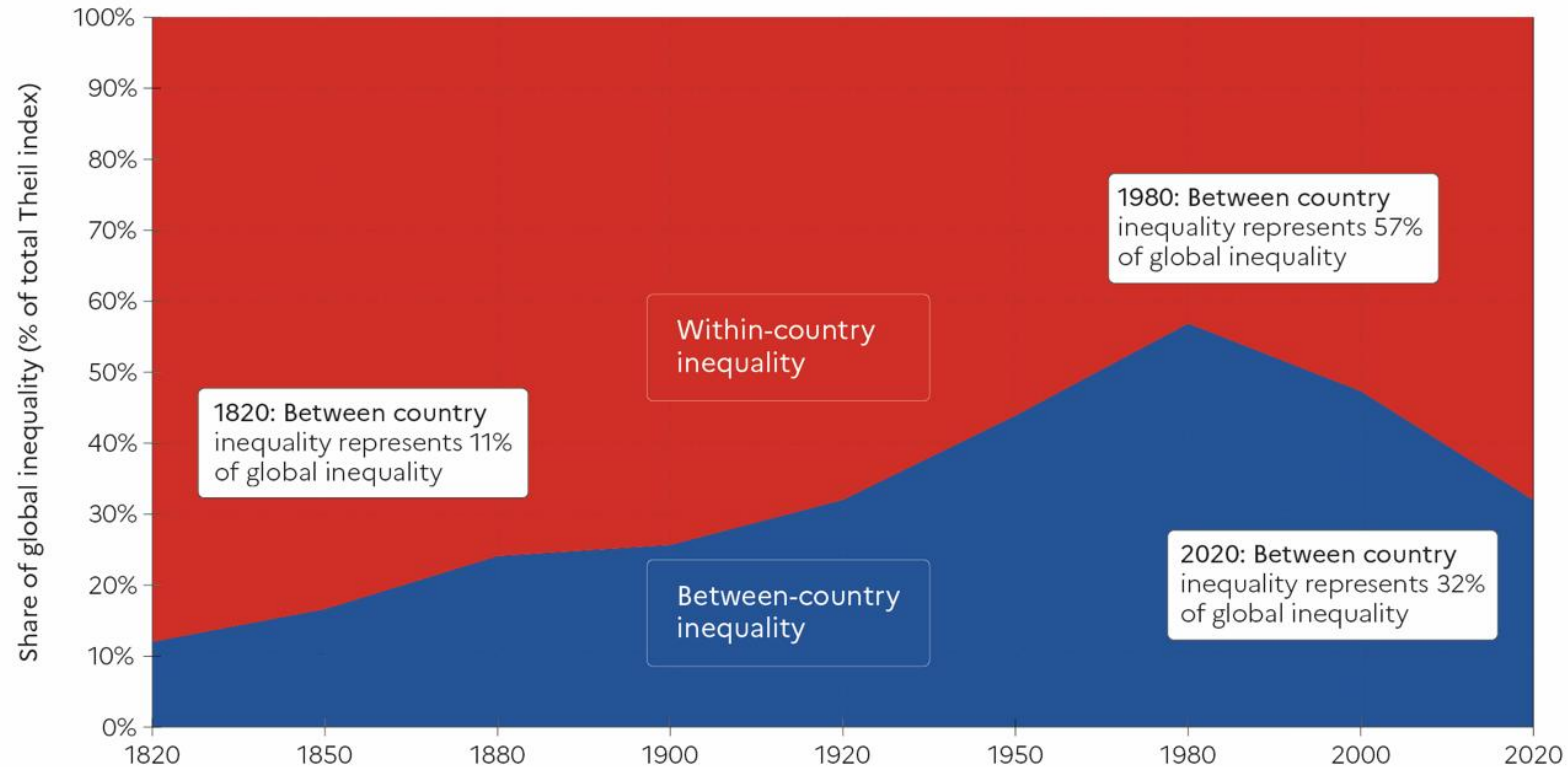


Interpretation: Global inequality, as measured by the ratio T10/B50 between the average income of the top 10% and the average income of the bottom 50%, more than doubled between 1820 and 1910, from less than 20 to about 40, and stabilized around 40 between 1910 and 2020. It is too early to say whether the decline in global inequality observed since 2008 will continue. Income is measured per capita after pension and unemployment insurance transfers and before income and wealth taxes. **Sources and series:** wir2022.wid.world/methodology and Chancel and Piketty (2021).

- Which inequality has increased in the last **20 years**?
- **Between-** or **Within-** Country Inequality?

But rising inequality within!

Figure 6 Global income inequality: Between vs. within country inequality (Theil index), 1820-2020

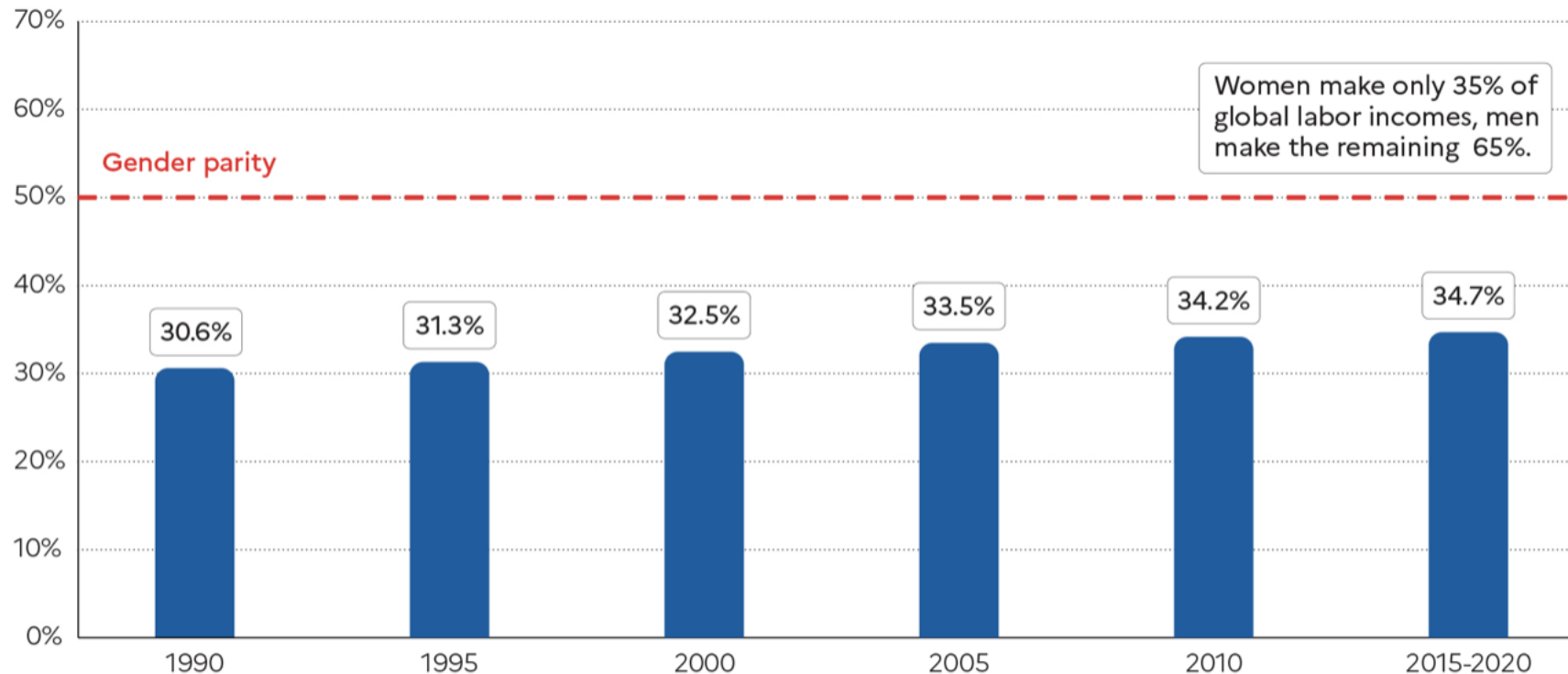


Interpretation: The importance of between-country inequality in overall global inequality, as measured by the Theil index, rose between 1820 and 1980 and strongly declined since then. In 2020, between-country inequality makes-up about a third of global inequality between countries. The rest is due to inequality within countries. Income is measured per capita after pension and unemployment insurance transfers and before income and wealth taxes. **Sources and series:** wir2022.wid.world/methodology and Chancel and Piketty (2021).

- What percentage of **labor income** is shared **globally** between **men/women**?

Within Inequality: Gender

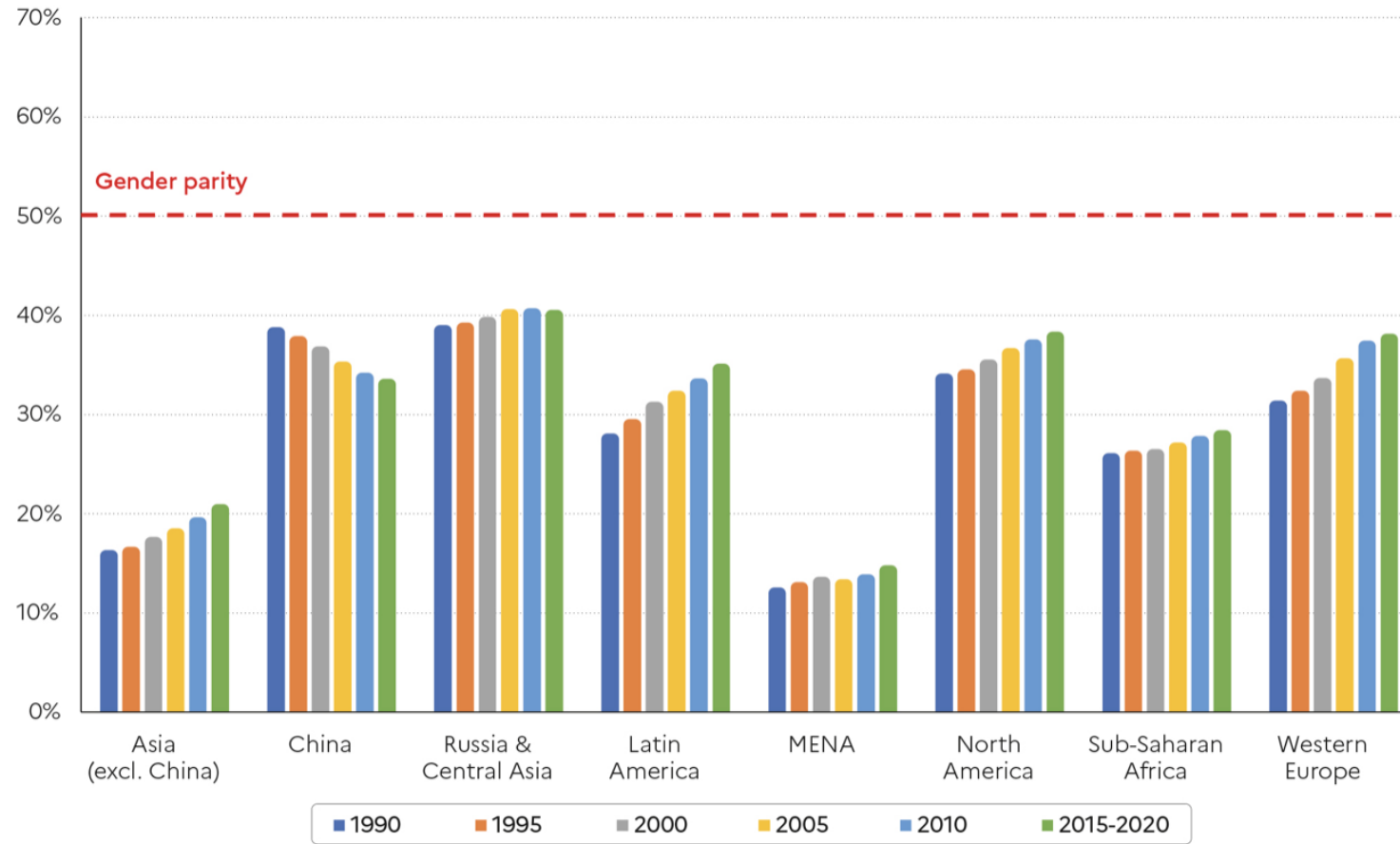
Figure 12 Female share in global labor incomes, 1990-2020



Interpretation: The share of female incomes in global labour incomes was 31% in 1990 and nears 35% in 2015-2020. Today, males make up 65% of total labor incomes. **Sources and series:** wir2022.wid.world/methodology and Neef and Robilliard (2021).

Even More Unequal in Other Regions...

Figure 13 Female labor income share across the world, 1990-2020



Interpretation: The female labour income share rose from 34% to 38% in North America between 1990 and 2020. **Sources and series:** wir2022.wid.world/methodology and Neef and Robilliard (2021).

1. Efficiency:

- Capital accumulation correlates with growth → Efficiency costs of capital taxation possibly greater

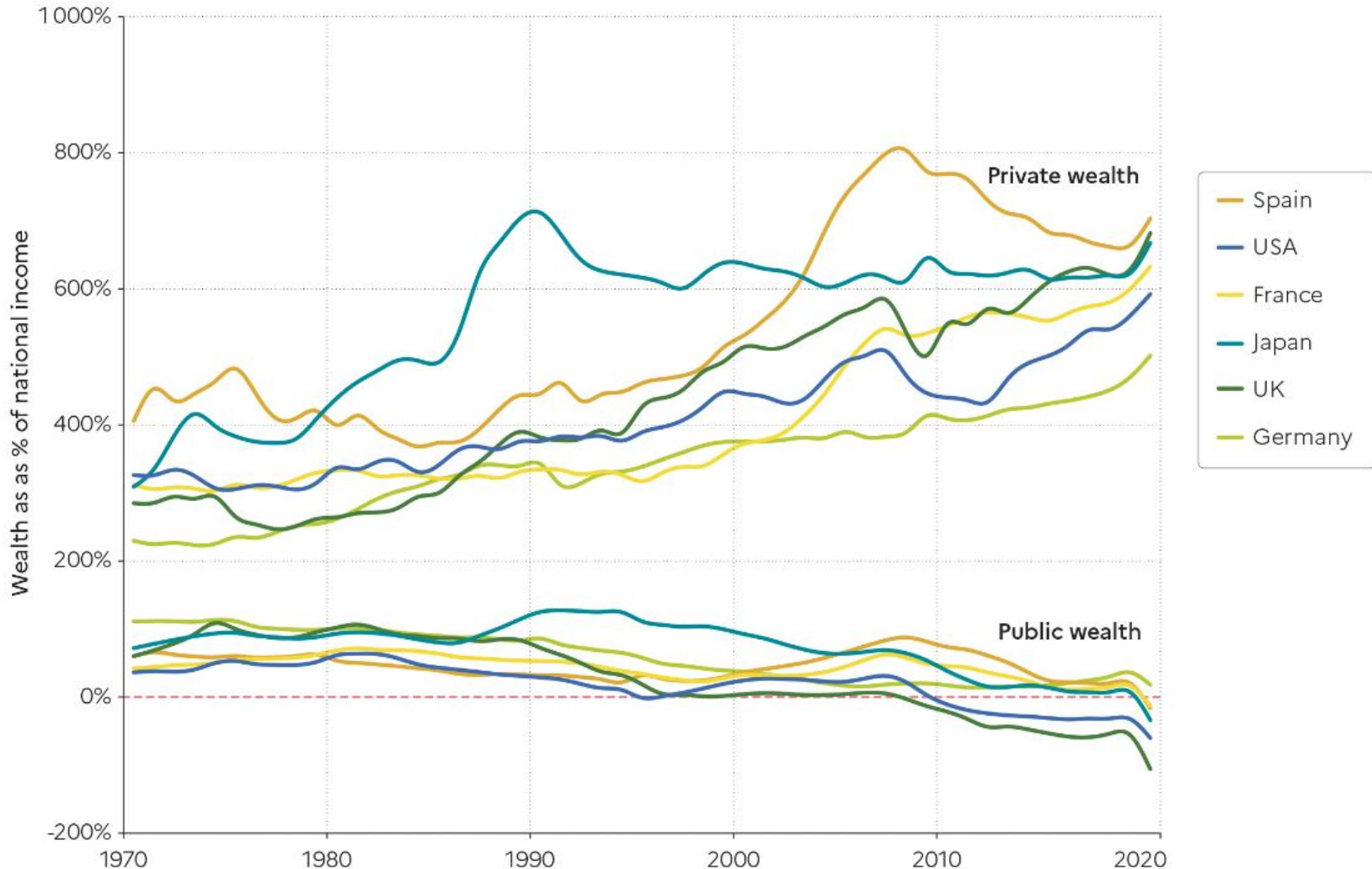
1. Redistribution:

- Capital more unequally distributed than income → Tax capital more than labor
- Capital income largely derived from inheritances → higher taxation (Equality of Opportunity)

1. Implementation:

- Capital is highly mobile internationally → Tax avoidance / -evasion
- Capital taxation is more complex than labor income taxation

Figure 8 The rise of private versus the decline of public wealth in rich countries, 1970-2020



Interpretation: Public wealth is the sum of all financial and non-financial assets, net of debts, held by governments. Public wealth dropped from 60% of national income in 1970 to -106% in 2020 in the UK. **Sources and series:** wir2022.wid.world/methodology, Bauluz et al. (2021) and updates.

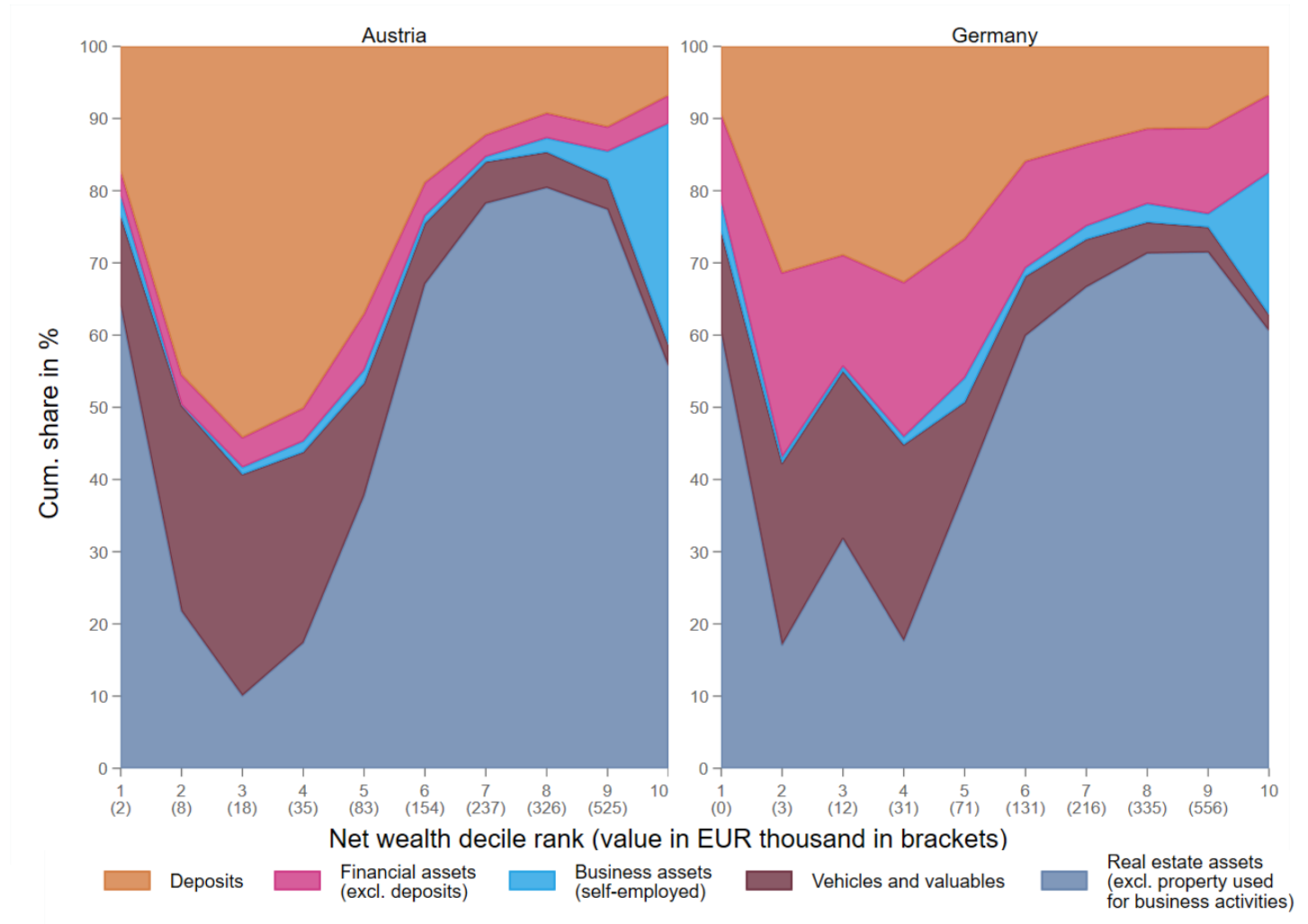
Warum Steigerung im privaten Kapital?

- Öffentliche Schulden (=Investitionen + Krisenmanagement)
- Privatisierungen
- Effizienz



What is private capital? I

- Components:
- Real Estate (Buildings and Land)
- Business Capital (or Corporate Capital)
- Durable Consumer Goods (e.g., Cars)
- Checking Accounts, Savings Accounts
- Other Financial Capital (Bonds, Stocks, etc.)
- less debts (or minus liabilities)



Intertemporal Consumption Decisions



Gruber, Kapitel 22

- **Consumption today vs. tomorrow:** Resources are limited – one must weigh up when consumption occurs.
- **Interest as the Price for Time:** Capital markets (or the "Savings Box + Interest") translate time into money.
- **Rich households** can invest & build wealth – **Poor** often cannot → **Inequality arises over time.**
- **Understanding political levers:** What effects do, for example, capital gains taxes, subsidies, and debt cancellation have on saving behavior and inequality?

Individual Decision I

- Individuals have differently high incomes between periods.
- The decision of how much to work has already been made → Income is exogenous.
- **Intuitively**: Employment + Pension.
- We want to know: **Consumption (Period 1,2), Savings Level, Utility**
- It is not optimal to consume the differently high incomes proportionally.
- Why?
- **Diminishing Marginal Utility!**
- Individuals distribute their income across consumption periods:
- Saving/Borrowing serves "consumption smoothing"

Model with capital market I

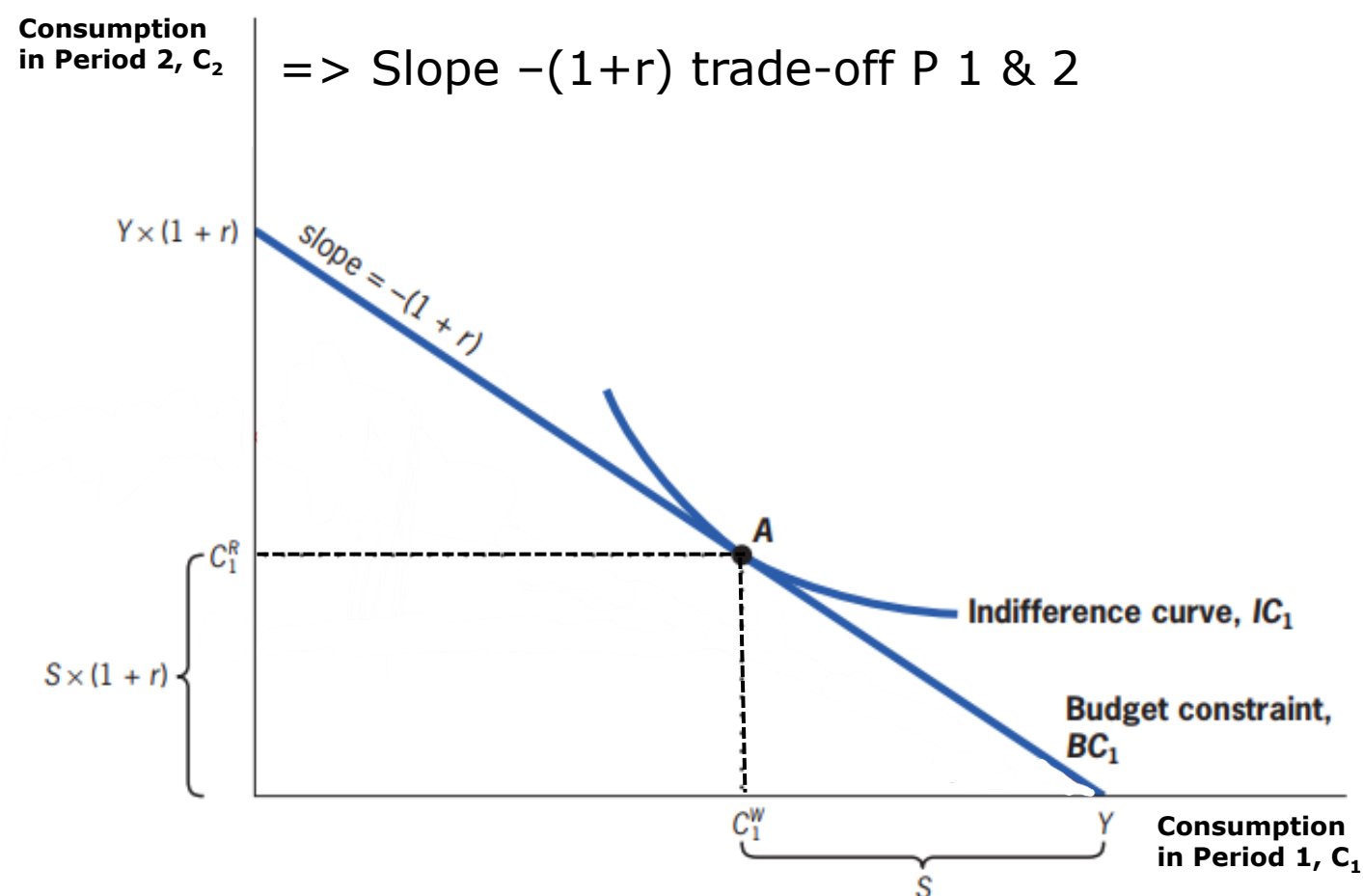
Assumptions:

- An individual **lives for 2 periods**: $t \in \{1, 2\}$
- In Period 1 income is Y_1 and in Period 2 Y_2 (**exogenously determined**)
- One cannot pass on the money → The entire income must be consumed after Period 2. The individual has a concave utility function
- The individual has a **concave utility function**
- There is a **capital market** – income can be freely allocated between the periods (**Borrowing, Saving is possible**)
- Interest rate r determines interest on savings and debts

Life-cycle model! – Assumption that wealth only arises from "life-cycle" saving

Model with capital market II

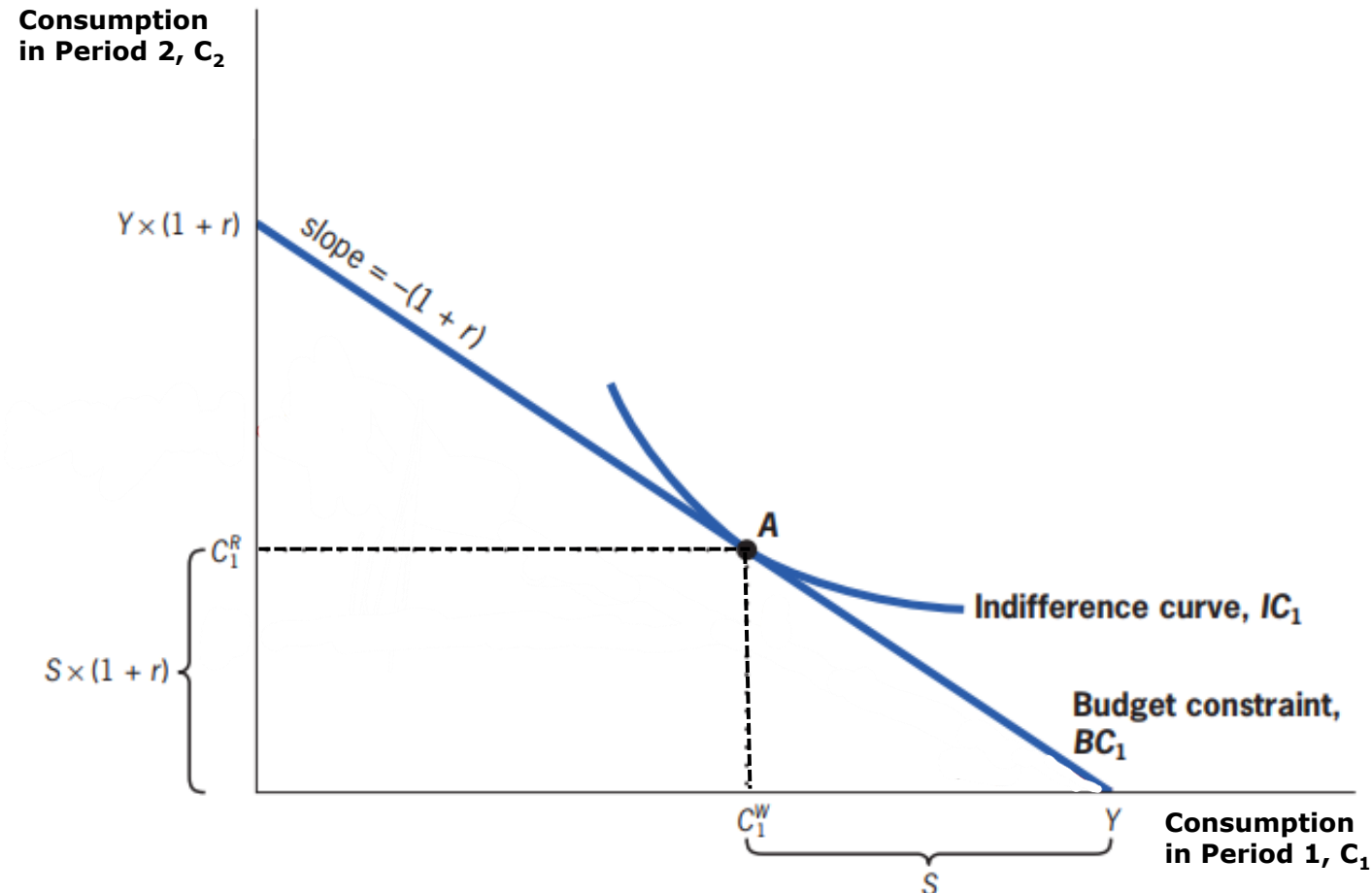
100% saved $\rightarrow Y \times (1+r)$ in Period 2



- Indifference Curve shows preference for current versus future consumption:
- Impatience
- How quickly the marginal utility of consumption declines
- Budget Line shows the relative price of current or future consumption (Interest Rate)

No saving $\rightarrow$ consumption Y in Period 1

Model with capital market III



- What is the Budget Constraint with a Capital Market?
- Consumption = Income (Note the different periods → Interest Rate!):

$$C_1 + \frac{C_2}{(1+r)} \leq Y_1 + \frac{Y_2}{(1+r)}$$

- maximize:

$$U = f(C_1, C_2)$$

- Subject to the budget constraint

$$C_2 \leq (Y_1 - C_1) * (1+r) + Y_2$$

$$\underbrace{\quad\quad\quad}_{S \times (1+r)}$$

Example I

An individual, who has a lifetime of 2 periods, has an income in Period 1 of 80€ (Y_1) and Period 2 von 20€ (Y_2). There is a capital market and an interest rate (r) = 20%. $U = \ln(C_1) + \ln(C_2)$
calculate C_1, C_2, S and U .

1st Step: Set up the Budget Line
(Budget Constraint)

Example I

1. step: Budget Constraint : $C_1 + \frac{C_2}{(1+r)} \leq Y_1 + \frac{Y_2}{(1+r)}$

$$C_1 + \frac{C_2}{(1,2)} \leq 80 + \frac{20}{(1,2)}$$
$$\frac{C_2}{(1,2)} \leq (80 - C_1) + \frac{20}{(1,2)}$$
$$C_2 \leq 116 - 1,2C_1$$

2nd Step: Substitute the Budget Line for C2 into the Utility Function and Differentiate with respect to C1

Example I

1. step: Budget Constraint : $C_1 + \frac{C_2}{(1+r)} \leq Y_1 + \frac{Y_2}{(1+r)}$

$$C_1 + \frac{C_2}{(1,2)} \leq 80 + \frac{20}{(1,2)}$$

$$\frac{C_2}{(1,2)} \leq (80 - C_1) + \frac{20}{(1,2)}$$

$$C_2 \leq 116 - 1,2C_1$$

2. 2nd Step: Substitute the Budget Constraint into the Utility Function:

$$U = \ln(C_1) + \ln(116 - 1,2C_1)$$

Solve for C_1 :

$$\frac{\partial U}{\partial C_1} = \frac{1}{C_1} - \frac{1,2}{116 - 1,2C_1} = 0$$

$$\rightarrow C_1 = 48,33$$

$$\rightarrow C_2 = 116 - 1,2C_1 = 58$$

$$\rightarrow S = Y_1 - C_1 = 80 - 48,33 = 31,67$$

$$\rightarrow U = \ln(C_1) + \ln(C_2) = 7,94$$

- Saving $\$ = Y_1 - C_1\$$
- Since money cannot be bequeathed (passed on), there is only saving in Period 1.
- Saving is modeled indirectly in this model!
- Similar to the labor market model, where we modeled leisure indirectly through working hours!

Mechanisms - Income

1. What happens to saving when income in Period 1 falls

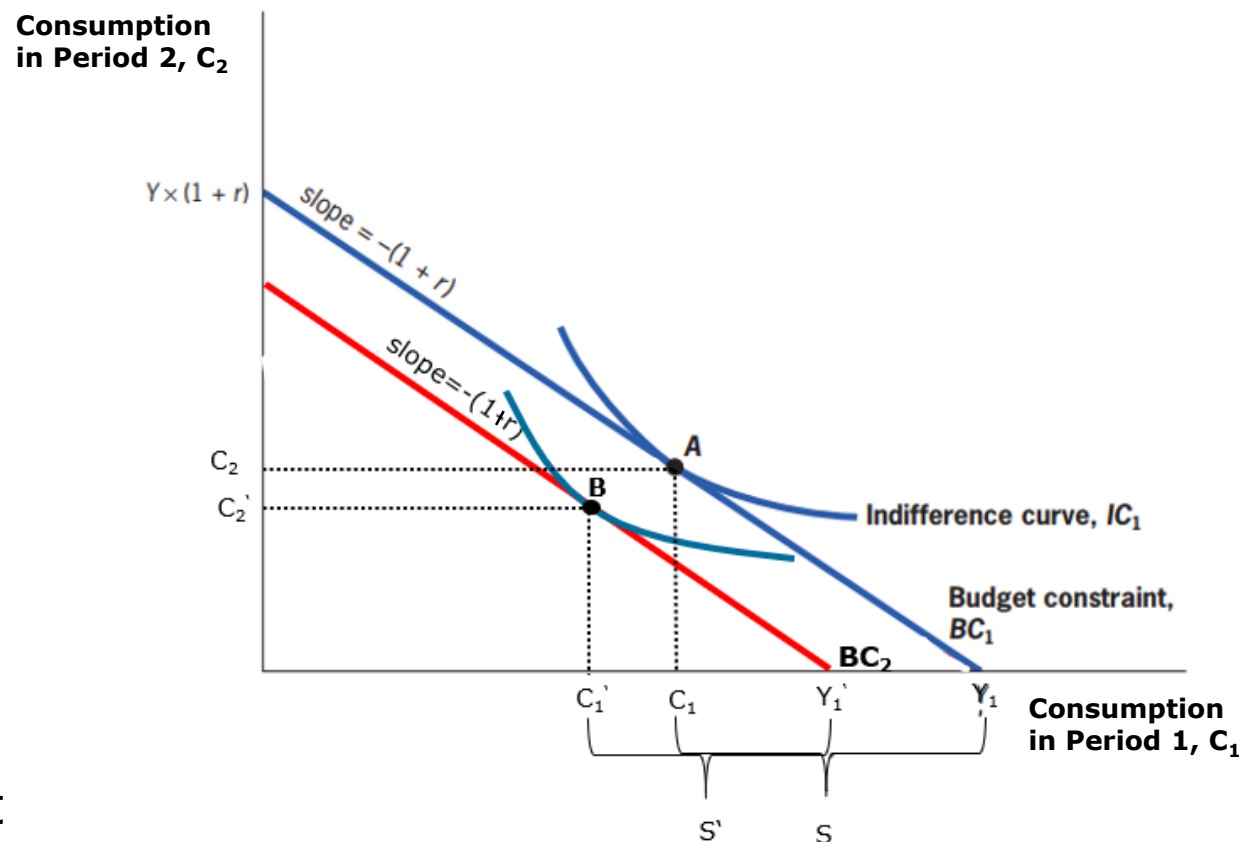
→ Is there a Substitution Effect?

→ No! The slope of the budget line and thus the relative price of consumption in Period 1 and 2 does not change. Only a parallel shift.

→ Is there an Income Effect

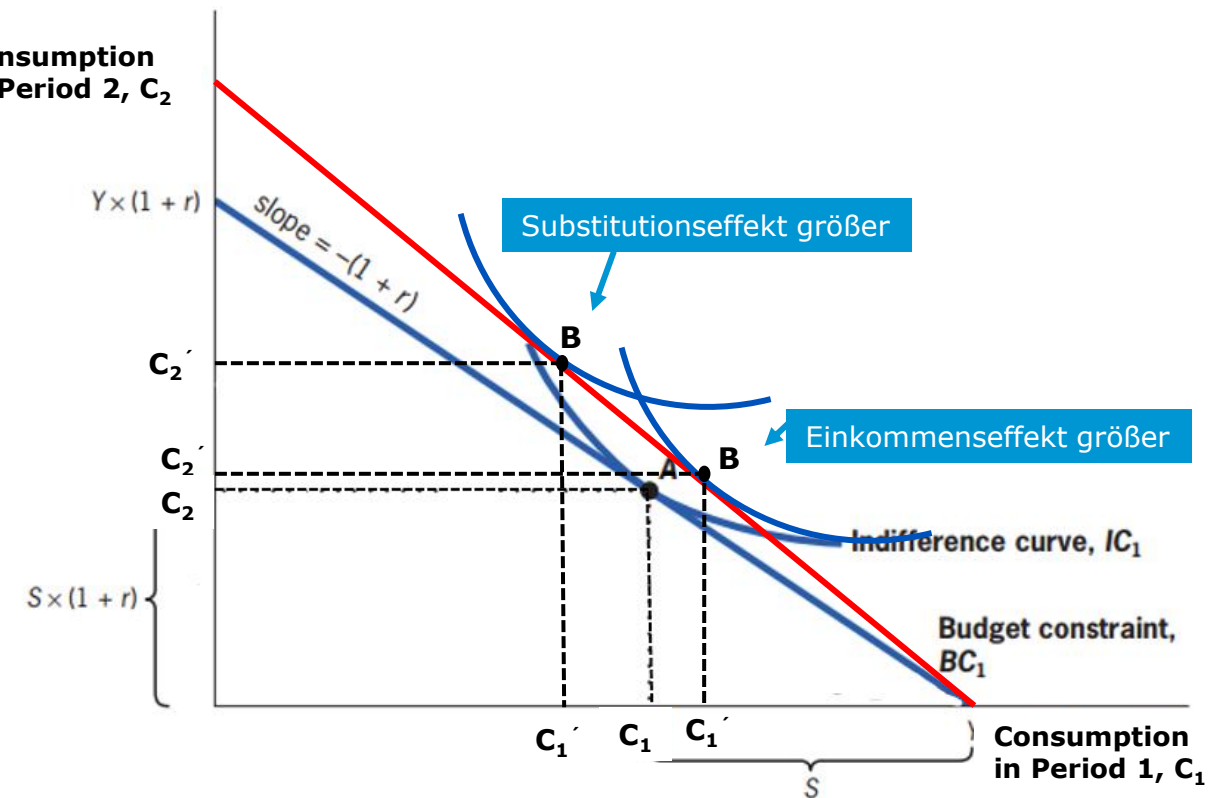
→ Yes. Less income available → less saving

- How does consumption change in Period 1 and in Period 2 → Both fall to the same extent



Mechanisms - Interest Rate

- What happens to saving when the interest rate increases → Is there a Substitution Effect?
- Yes! Saving is now more worthwhile because interest rates are higher → **Substitution from consumption in Period 1** to saving (and consumption in Period 2)
- Is there an Income Effect
- Yes! Because with the same amount of saving, the interest is higher, less saving is now required to maintain consumption in Period 2 → Saving falls and consumption in Period 1 rises
- Think, for example, of a savings goal for a pension How does consumption change in Period 1 and in Period 2 → Unclear...



Taxation in the 2-Period Model

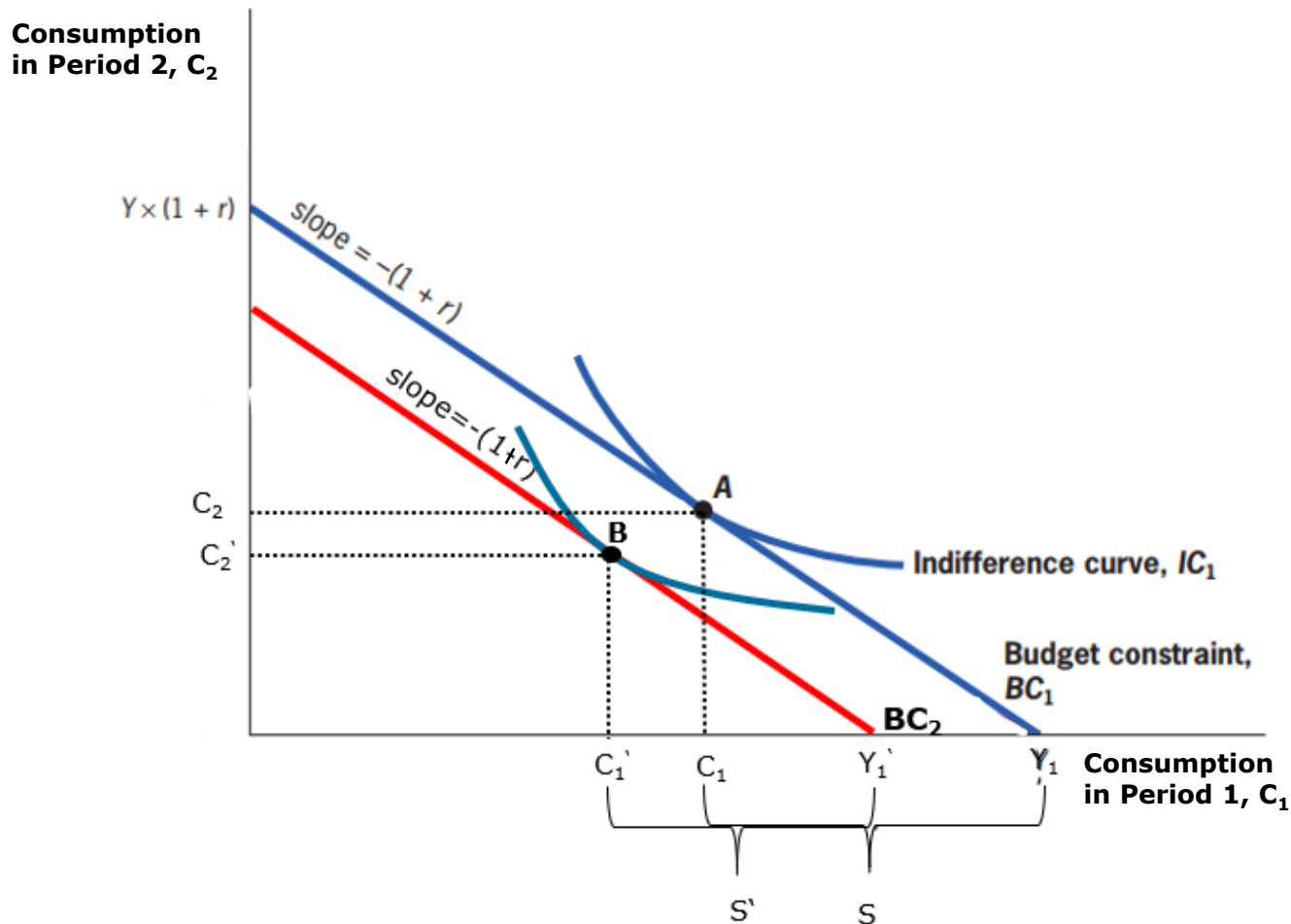


Gruber, Kapitel 22

Points of Taxation in the Model

- **Income Generation → Income Tax**
- Taxes on the Increase in Wealth (Capital Growth) → Capital Gains Tax

Income Tax



What is the **Budget Constraint with Income Tax**?

$$C_1 + \frac{C_2}{(1 + r)} \leq Y_1(1 - \tau_Y) + \frac{Y_2}{(1 + r)}(1 - \tau_Y)$$

τ_Y

→ Budget constraint shifts parallelly

max:

$$U = f(C_1, C_2)$$

S.t

$$C_2 \leq (Y_1(1 - \tau_Y) - C_1) * (1 + r) + Y_2(1 - \tau_Y)$$

Example II

Assume an individual who has a lifetime of **2 periods** has an income in Period 1 of 80€ (Y_1) and Period 2 of 20€ (Y_2). There is a capital market and the interest rate (r) = 20%. $U = \ln(C_1) + \ln(C_2)$

a) calculate C_1, C_2, S and U .

→ Calculate *example I*: $C_1 = 48,33$, $C_2 = 58$, $S = 31,67$, $U = 7,94$

b) Assume the government introduces an **Income Tax of 25%** ($Y_n = Y_b(1 - \tau_Y)$). How do your answers to a) change?

Example II

b) 1. New Budget Constraint: $C_1 + \frac{C_2}{(1+r)} \leq Y_1(1 - \tau_Y) + \frac{Y_2}{(1+r)}(1 - \tau_Y)$

$$C_1 + \frac{C_2}{(1,2)} \leq 80(1 - 0,25) + \frac{20}{(1,2)}(1 - 0,25)$$

$$C_2 \leq 87 - 1,2C_1$$

Slope remains the same! No Substitution Effect

2. Substitute into Utility Function::

$$U = \ln(C_1) + \ln(87 - 1,2C_1)$$

Derive wrt C1:

$$\frac{\partial U}{\partial C_1} = \frac{1}{C_1} - \frac{1,2}{87 - 1,2C_1} = 0$$

Calculated with Income after Taxes!

$$\rightarrow C_1 = 36,25$$

$$\rightarrow C_2 = 87 - 1,2C_1 = 43,5$$

$$\rightarrow S = 80(1 - 0.25) - 36,25 = 60 - 36,25 = 23,75$$

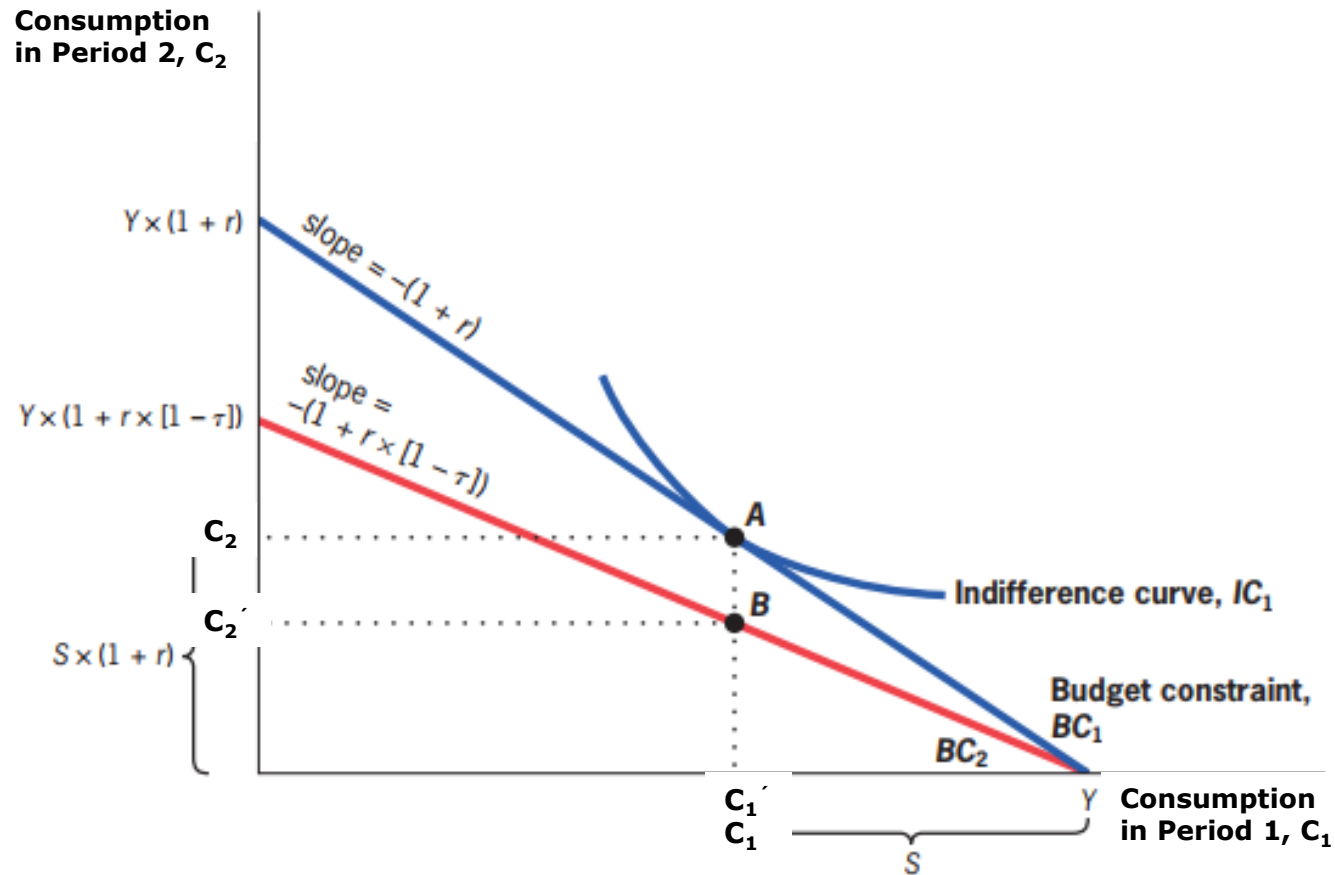
$$\rightarrow U = 7,36$$

- How does the consumption decision change when the government introduces an income tax of 25%?
- Does the Indifference Curve change?
- → NO!
- Does the Budget Constraint change?
- → YES!
- Does Saving change?
- → Less is saved. → Income Effect
- Does the relationship between C_1 and C_2 change?
- → No, the slope of the budget line does not change!
- → Only Income Effect, no Substitution Effect

Points of Taxation in the Model

- **Income Generation → Income Tax**
- Taxes on the Increase in Wealth (Capital Growth) → Capital Gains Tax
- Taxes on the Stock of Wealth → Wealth Tax

Taxes on capital



What is the budget constraint with taxes on capital?

$$C_1 + \frac{C_2}{(1 + r(1 - \tau_c))} \leq Y_1 + \frac{Y_2}{(1 + r(1 - \tau_c))}$$

→ budget constraint shifts downwards

max:

$$U = f(C_1, C_2)$$

S.t.

$$C_2 \leq (Y_1 - C_1) * (1 + r(1 - \tau_c)) + Y_2$$

Example II

Assume an individual who has a lifetime of **2 periods** has an income in Period 1 of 80€ (Y_1) and Period 2 of 20€ (Y_2). There is a capital market and the interest rate (r) = 20%. $U = \ln(C_1) + \ln(C_2)$

a) calculate C_1, C_2, S and U .

→ *Siehe Bsp I*: $C_1 = 48,33$, $C_2 = 58$, $S = 31,67$, $U = 7,94$

b) Assume the government introduces an **Income Tax of 25%** ($Y_n = Y_b(1 - \tau_Y)$). How do your answers to a) change?

→ *see Bsp IIb*: $C_1 = 36,25$, $C_2 = 43,5$, $S = 23,75$, $U = 7,36$

c) Assume the government introduces a capital gains tax of 25% ($r_n = r_b(1 - \tau_c)$). How do your answers change?

Example II

c) 1. New Budget Constraint : $C_1 + \frac{C_2}{(1+r(1-\tau_C))} \leq Y_1 + \frac{Y_2}{(1+r(1-\tau_C))}$

$$C_1 + \frac{C_2}{(1 + 0,2(1 - 0,25))} \leq 80 + \frac{20}{(1 + 0,2(1 - 0,25))}$$

$$C_2 = 112 - 1,15C_1$$

Substitute into Utility Function:

$$U = \ln(C_1) + \ln(112 - 1,15C_1)$$

Derive wrt C1:

$$\frac{\partial U}{\partial C_1} = \frac{1}{C_1} - \frac{1,15}{112 - 1,15C_1} = 0$$

$$\rightarrow C_1 = 48,70$$

$$\rightarrow C_2 = 112 - 1,15C_1 = 56$$

$$\rightarrow S = 80 - 48,70 = 31,3$$

$$\rightarrow U = \ln(48,70) + \ln(56) = 7,91$$

Slope becomes flatter!

Capital Gains Tax IV

- How does the consumption decision change when the government introduces a Capital Gains Tax of 25%?
- Does the Indifference Curve change?
- →NO!

- Does the **Budget Constraint** change?
- YES, shifts inwards!
- Does **Saving** change?
- Not clear.

How do the Income and Substitution Effects work when a Capital Gains Tax is introduced?

How does a Capital Gains Tax affect saving behavior?

- **Substitution Effect:**

- Saving becomes more expensive → less is saved and consumption in Period 2 decreases
- The opportunity cost of consumption in Period 1 falls because one "gets less out" of saving in Period 2

- **Income Effect:**

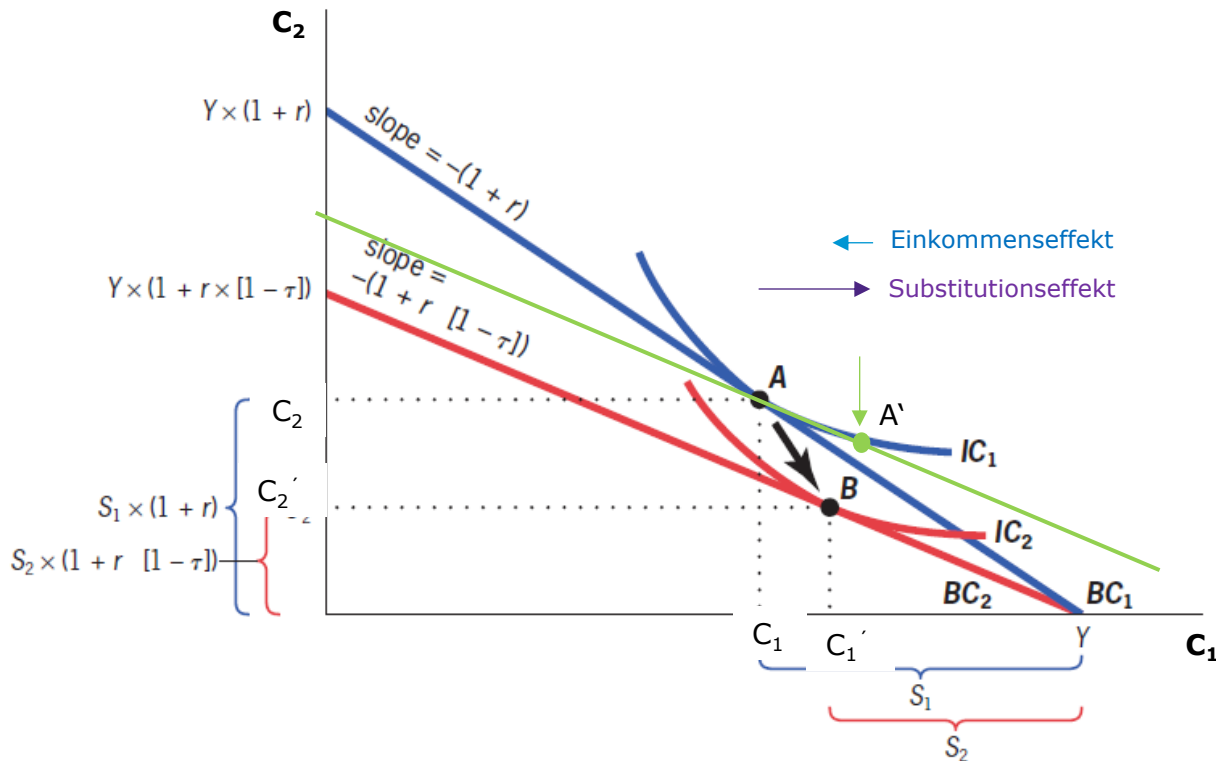
- Saving increases to secure the consumption level in Period 2. Consumption in Period 1 decreases.
- To compensate for the lower interest payments for saving, more must be saved.
→ Which effect dominates is not clear ex-ante.
- → Also empirically difficult to measure – evidence suggests the Substitution Effect is larger, but minimal

Capital Gains Tax IV

- How does the consumption decision change when the government introduces a Capital Gains Tax of 25%?
- Does the Indifference Curve change
- →NO!
- Does the Budget Constraint change?
- →YES, shifts inwards!
- →Does Saving change?
- → Yes, it decreases! →Substitution Effect dominates
- Does the relationship between C1 and C2 change?
- $\rightarrow$ Yes! C1 rises in relation to C2(Substitution Effect dominates)

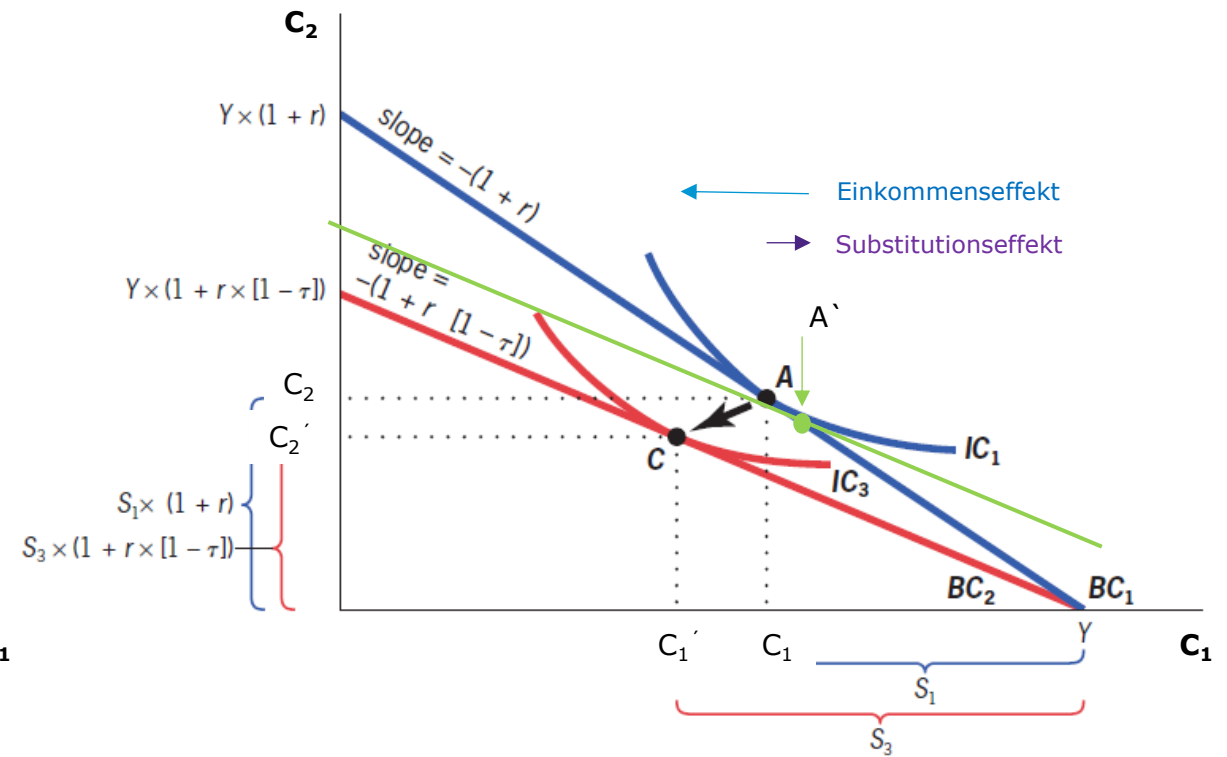
Where is the substitution and the income effect?

(a) Substitution effect is larger



→ C_2 Decreases more than C_1

(b) Income effect is larger



→ C_1 Decreases more than C_2

Optimal taxation of the Factor markets-Theory



Emmanuel Saez: *Undergraduate Public Economics* (<https://eml.berkeley.edu/~saez/course131/course131.html>)

Stiglitz, Rosengard (2015) Kap.21

OECD (2018), *The Role and Design of Net Wealth Taxes in the OECD*, OECD Tax Policy Studies, No. 26, OECD Publishing, Paris, <https://doi.org/10.1787/9789264290303-en>.

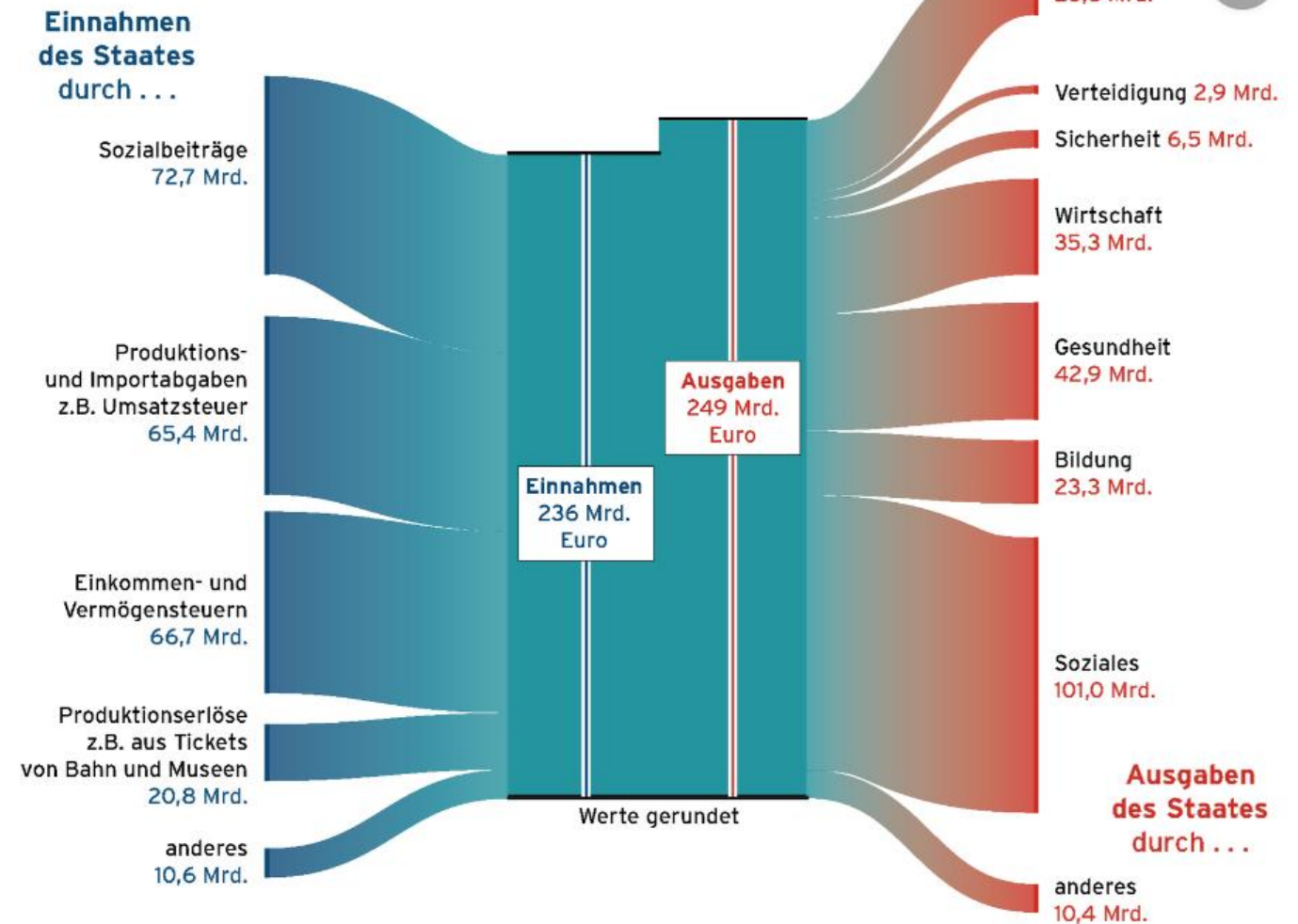
What speaks in favor of capital taxation?

- Pro-Argument
- Contra-Argument

- Reduce wealth inequality and promote equal opportunity:
- Self-reinforcing wealth accumulation;
- Power & political influence → implications for democracy;
- Inheritances hinder social mobility and contradict the merit principle.
- Capital income and labor income are not always clearly distinguishable → unequal tax treatment leads to tax shifting and inefficiencies.
- Wealth generates advantages beyond capital income → a capital income tax alone is not sufficient.
- No sacrifice of leisure is required → performance/effort-based argument.
- Wealth brings power, social status, opportunities, security, life satisfaction, etc.

■ Budget-Deficit!!

Österreichs Einnahmen und Ausgaben 2023



- [Österreichs Staatsausgaben: So geht es besser! | Der Pragmaticus](#)

- **Double Taxation:**
- Difficult in relation to income tax → often the case (e.g., taxed income and VAT upon purchase)
- **Distortion of Savings and Investment Behavior**
- Investment incentives decrease (very elastic production factor, see Block 2)
- Taxation of all future uses of capital → more inefficiencies
- BUT: Inheritance tax could even stimulate productive use of assets
- **Negative Incentives for Entrepreneurship and Willingness to Take Risks**
- Dependent on the treatment of losses and correlation to income
- Business assets are often excluded/exemptions are often granted
- **Liquidity Problems with Illiquid Assets → Sales**

- **Tax Evasion/Avoidance: "Voluntary Tax"**
 - Intensified by more **Exemption Rules** → Higher exemptions, but for a broad tax base (no exceptions or privileged asset classes)
 - Easier for **Inheritance Taxes** (only one point in time)? (Scheuer, Slemrod 2021)
 - **Global Question?** → International exchange of information and taxation of emigrants (a few years)
- **Valuation of assets** (easier for inheritance tax because the question of regularity and the time point is eliminated)
- Large portion of top assets (Zucman: 70%) have **clear market values** (financial assets, real estate, etc.)
- Some types of assets are more difficult (e.g., Art)
- Modern Technologies: Data on market values of assets are easier to calculate → **not self-reported!**

Optima capital taxation I

Traditionally: 0% tax on capital!

- Atkinson, A. B. and J. E. Stiglitz. 1976. "The design of tax structure: direct versus indirect taxation." *Journal of Public Economics* 6: 55-75.
- Barro, Robert J. 1974. "Are Government Bonds Net Wealth?" *The Journal of Political Economy*: 82(6): 1095-1117.
- Chamley, C. (1986). "Optimal taxation of capital income in general equilibrium with infinite lives." *Econometrica* 54(3): 607–622.
- Diamond, Peter A. and James A. Mirrlees. 1971. "Optimal Taxation and Public Production I: Production Efficiency," *American Economic Review* 61(1): 8-27.
- Judd, K. L. 1985. "Redistributive taxation in a simple perfect foresight model." *Journal of Public Economics* 28(1): 59–83.
- Ramsey, Frank. 1927. "A Contribution to the Theory of Taxation," *Economic Journal* 37 (145): 47-61.

Atkinson-Stiglitz' theorem

- **Atkinson-Stiglitz' theorem:** Optimal tax on capital income: **0% !**
 - => Taxes on labor are sufficient!
- **Intuition:** In the Life-Cycle Model, **Inequality** arises **only** due to **differences in earning capacity..**
 - Taxes on labor can minimize these inequalities.
 - Taxing capital income unnecessarily distorts savings behavior.
- Does it seem fair not to discriminate against savers if labor income is the sole source of inequality?

Optimal capital taxation II

Mankiw et al. (JEP, 2009)

- 0% capital tax

Diamond and Saez (JEP, 2011)

- Positive capital tax

The precise assumption of the models is decisive (e.g., planning horizon of the actors, savings rate)! Results are not unambiguous.

Optimal capital taxation III

capital income

- Capital income
 - Tax on capital income distorts investment and savings decisions and thus harms the national economy more than taxes on labor income or consumption → Optimal 0% (or less than the tax rate on labor income)!
 - **BUT:**
 - **Distinction** between labor and capital income is difficult: "**tax shifting**" when capital income is taxed less (Finland: Pirttilä, Selin (2011))
 - **Wealthy** households often derive their income mainly from **capital** → would otherwise be entirely untaxed
- (At least) the same tax rate as for labor income!

Optimal capital taxation IV

Assets

- **Assets** are created from saving during lifetimes (**Life-cycle model**):
- Capital income tax/wealth tax distorts savings behavior and "punishes" those who want to save more
- **Wealth inequality** arises from differences in **income** and different saving decisions → income inequality is better addressed with income tax
- → **optimal 0%!**
- **BUT:**
 - **Different returns** (over time and between people - more wealth, higher returns) → **Wealth Tax >0%!**
 - Assets from inheritances (**inherited wealth**) → **Wealth Tax >0%!**

Optimal capital taxation V

Inheritances

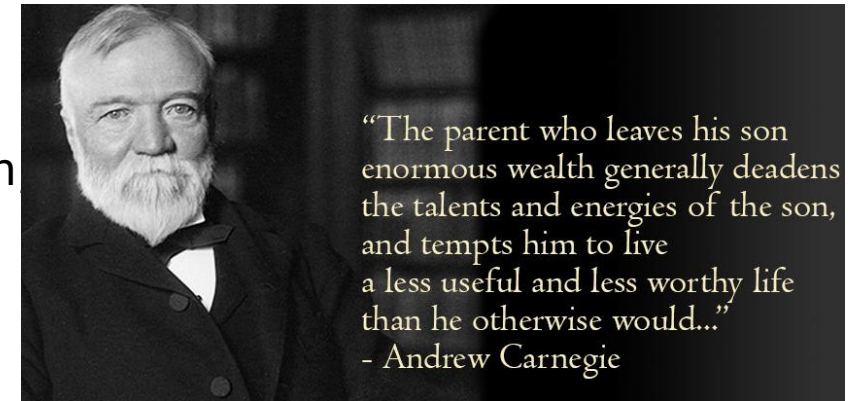
- **Why are there inheritances?** (empirically unclear – Kopczuk 2013)

- **Unintentional:**

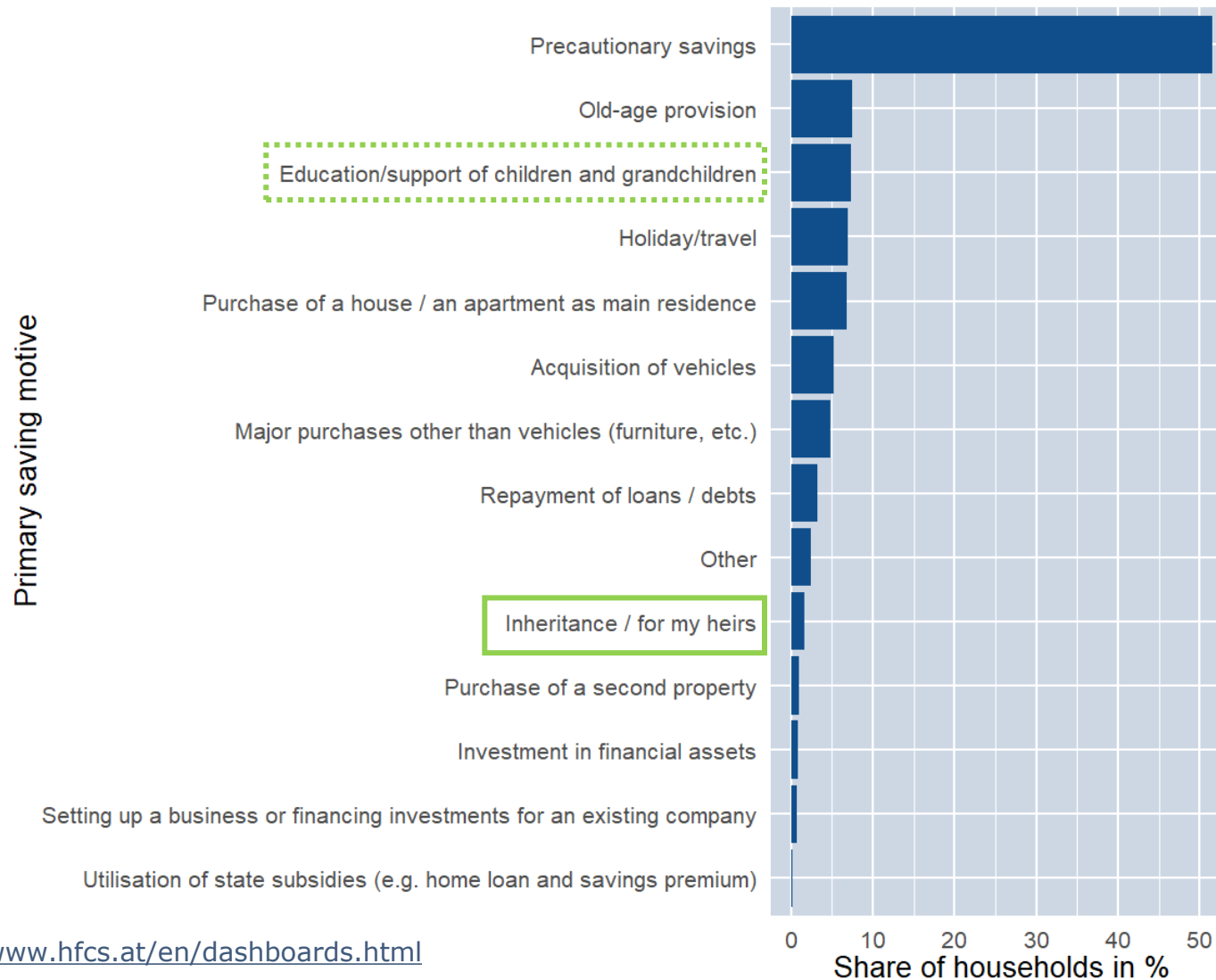
- No distortions (because not planned),
- but lower **labor supply of heirs** (Holtz-Eakin, Joulfaian Rosen 1993)

→ **High tax rates optimal!**

- Altruistic/Social Motives:
- Inheritance is planned (saved for) and brings utility for the heir and the benefactor
- tax reduces wealth accumulation and labor supply of altruistic parents (and utility)
→ **Lower/No tax rates**



Reasons for saving in Austria



Political Implications



Fessler et al. (2017); HFCS

OECD-Report: „The Role and Design of Net Wealth Taxes in the OECD“

Gruber, Kapitel 23

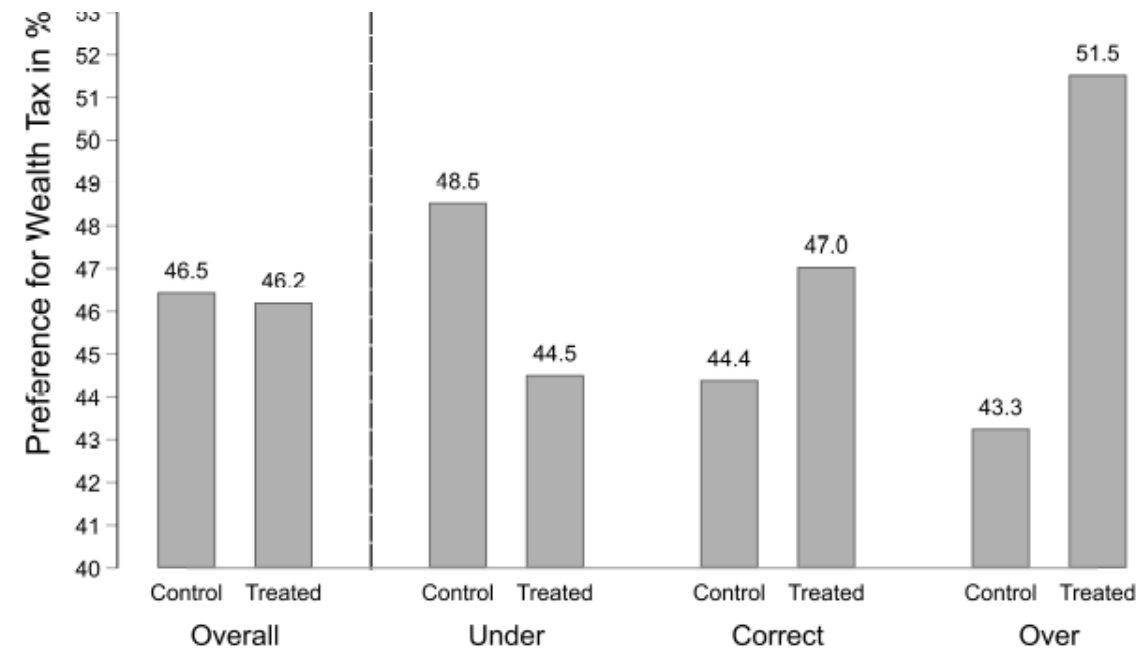
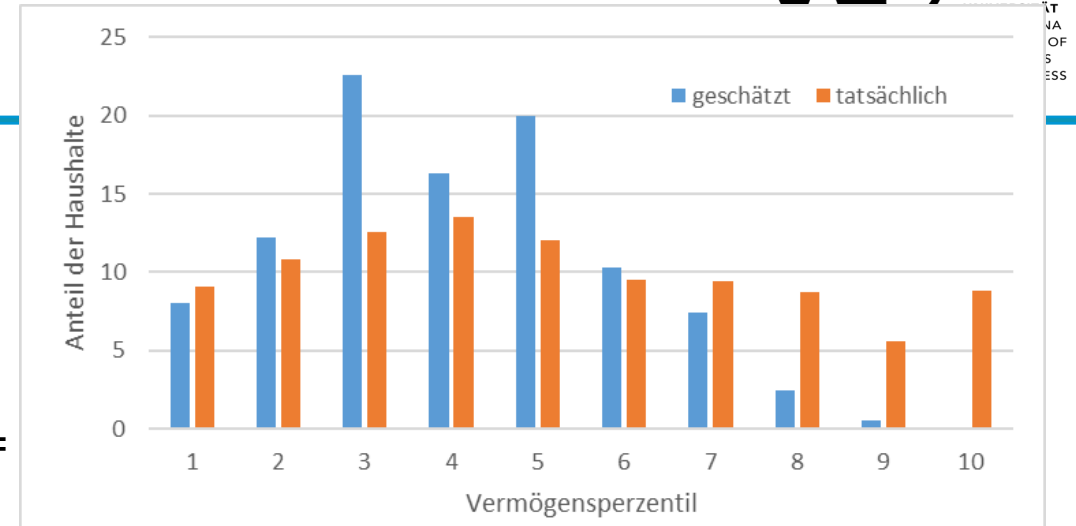
Gabriel Zucman Lecture: Public Policy (<https://gabriel-zucman.eu/pp290/>)

Emmanuel Saez Lecture: Undergraduate Public Economics (<https://eml.berkeley.edu/~saez/course131/course131.html>)

<https://wir2022.wid.world/chapter-7/>

Albacete et al (2022) The Wealth Distribution and Redistributive Preferences

- Experiment in HFCS in Austria:
 - Question about own position in the wealth distribution
 - 50% receive information about their position (= Treatment)
 - Questions on taxes and redistribution
- Results:
 - Many households **incorrectly estimate their own position**
 - Could affect support for tax!!

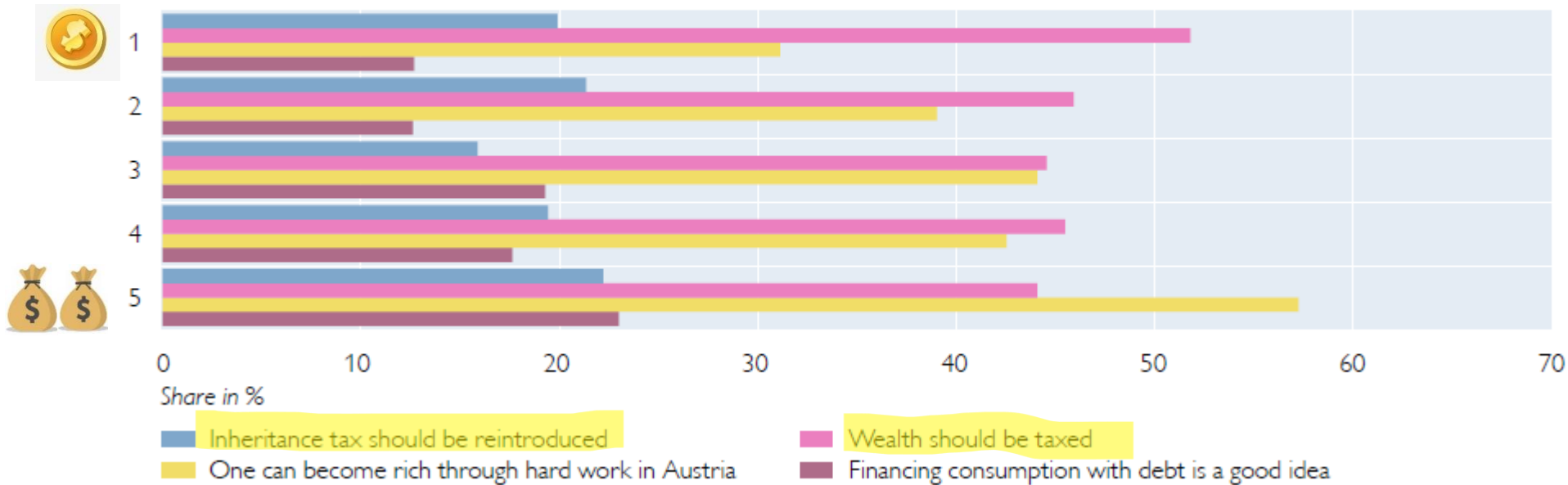


Survey Results

Chart 3

Views and preferences of the financially knowledgeable person across net wealth quintiles

Net wealth quintiles



Source: HFCS Austria 2017, OeNB.

Empirical Evidence



Kapitaleinkommenssteuer

Enough empirical evidence, Bsp: Yagan (2015) *Capital Tax Reform and the Real Economy: The Effects of the 2003 Dividend Tax Cut*

- Analyzed the **reduction of the capital gains tax** in the U.S.A. in 2003 on investments and labor income:
- Bush: reduction from 38,5% to 15%
 - C-corporations affected by tax cut, comparable S-corporations not
- Finds no clear effects
- **BUT Dividend payouts to shareholders** clearly increase

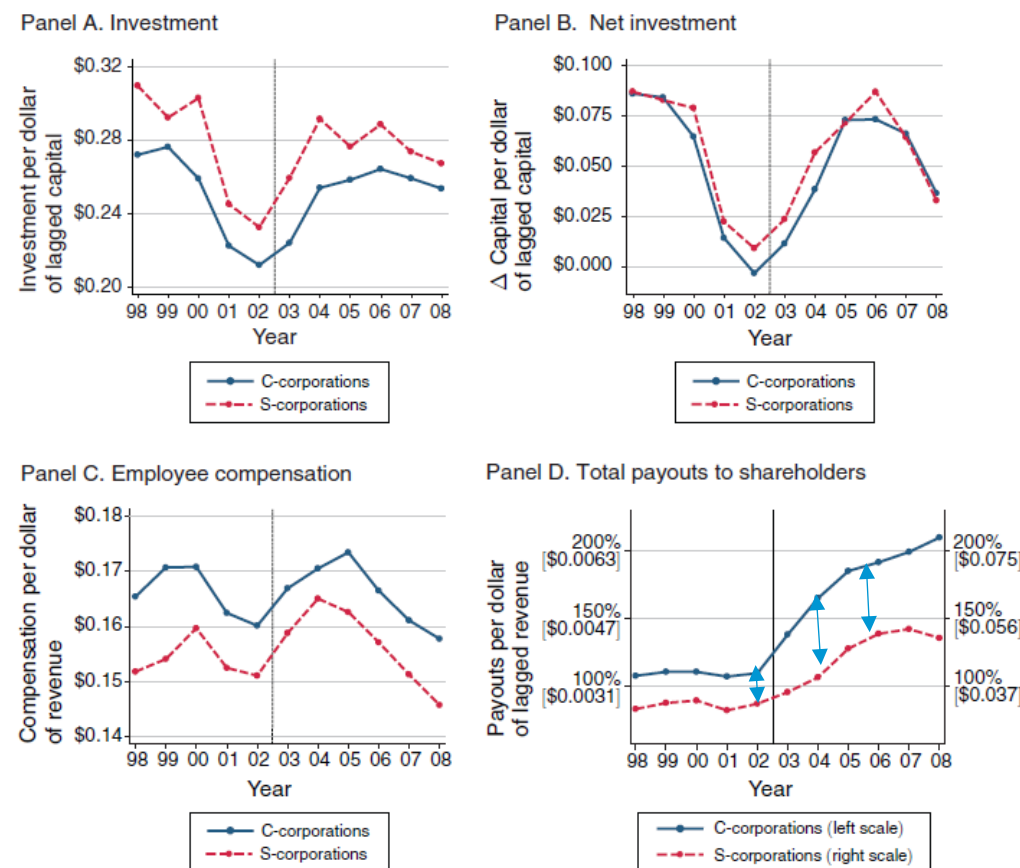


FIGURE 2. EFFECTS OF THE 2003 DIVIDEND TAX CUT

Marti, S., Martínez, I. Z., & Scheuer, F. (2023). Does a progressive wealth tax reduce top wealth inequality? Evidence from Switzerland. *Oxford Review of Economic Policy*, 39(3), 513-529.

- **Uses Variation in wealth tax rates** across time and cantons in Switzerland:
- Tax clearly has an effect on distribution: **Increases reduce, Reductions increase wealth inequality**
- A **0.1%-point reduction** in the tax rate increases the **Top 1%(0.1%) share by 0.9(1.2) %-points** five years later
- Tax reductions of the last 50 years can explain **18%(25%) of the rising share of the Top 1%(0.1%)**

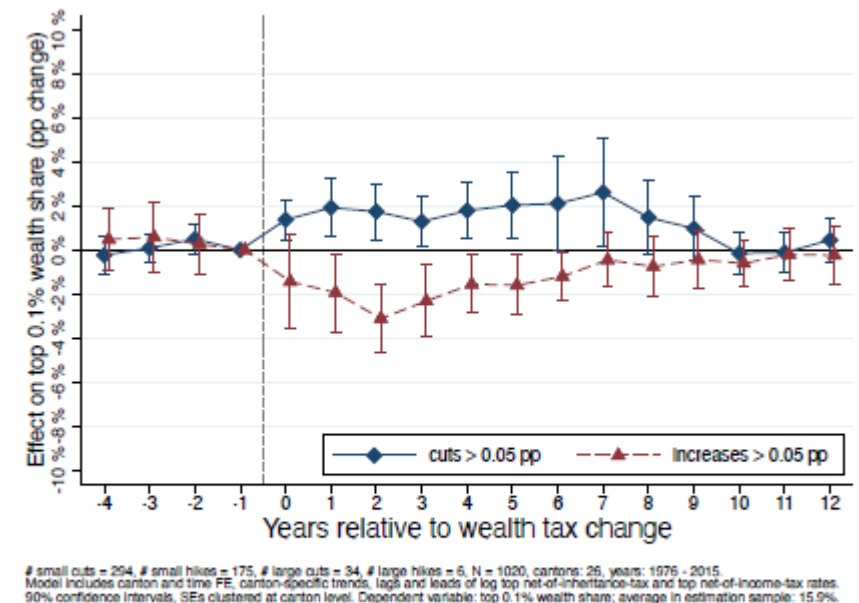


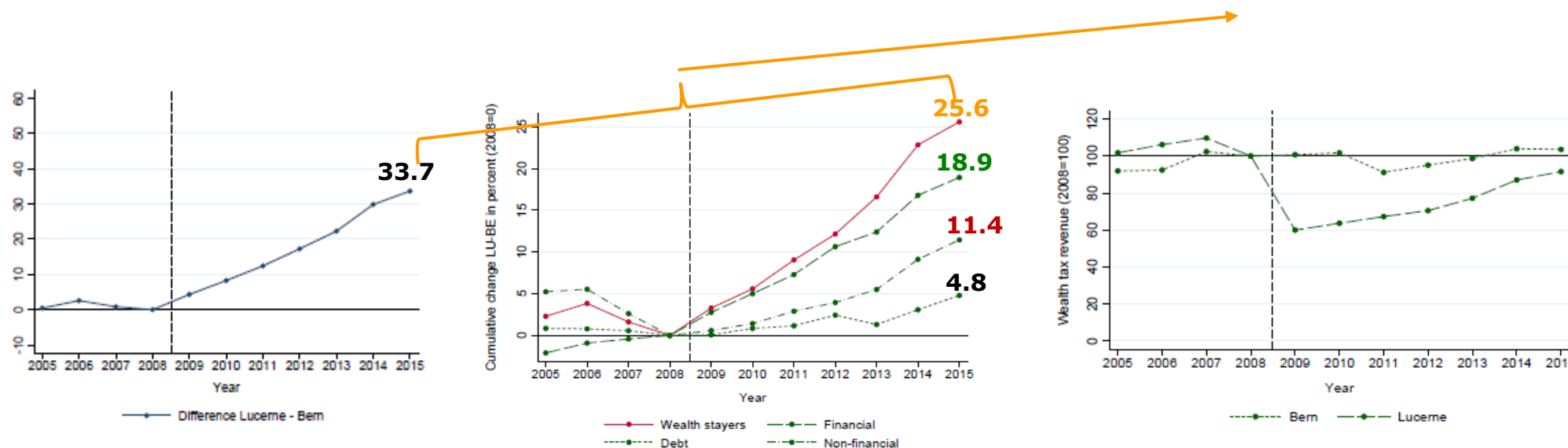
Figure 8: Cross-canton event study, top 0.1% wealth share

How Mobile is Capital?

- Income rather **inelastic** (Saez, Slemrod, & Giertz 2012)
- **Reasons for Mobility:**
 - Assets (in contrast to labor income) are in the "control" of the employees.
 - Asset monitoring is more difficult for authorities (taxation).
- **Reasons against Mobility:**
 - Most of the capital is in immobile form – houses.
 - Difficult to adjust in the short term!

Brülhart et al (2022) Behavioral responses to wealth taxes – evidence from Switzerland

- Context: Lucerne reduced its tax rate by 50% in 2009, with Bern serving as the control group with only a 14% reduction
- Result: Wealth accumulation in Lucerne is 33.5 percentage points (!) higher than in Bern (stronger reaction with higher wealth)
- Decomposition: 6% increased savings/mechanical, 24% Migration, 21% House prices, 49% increase in the value of financial assets (→previously underreported)

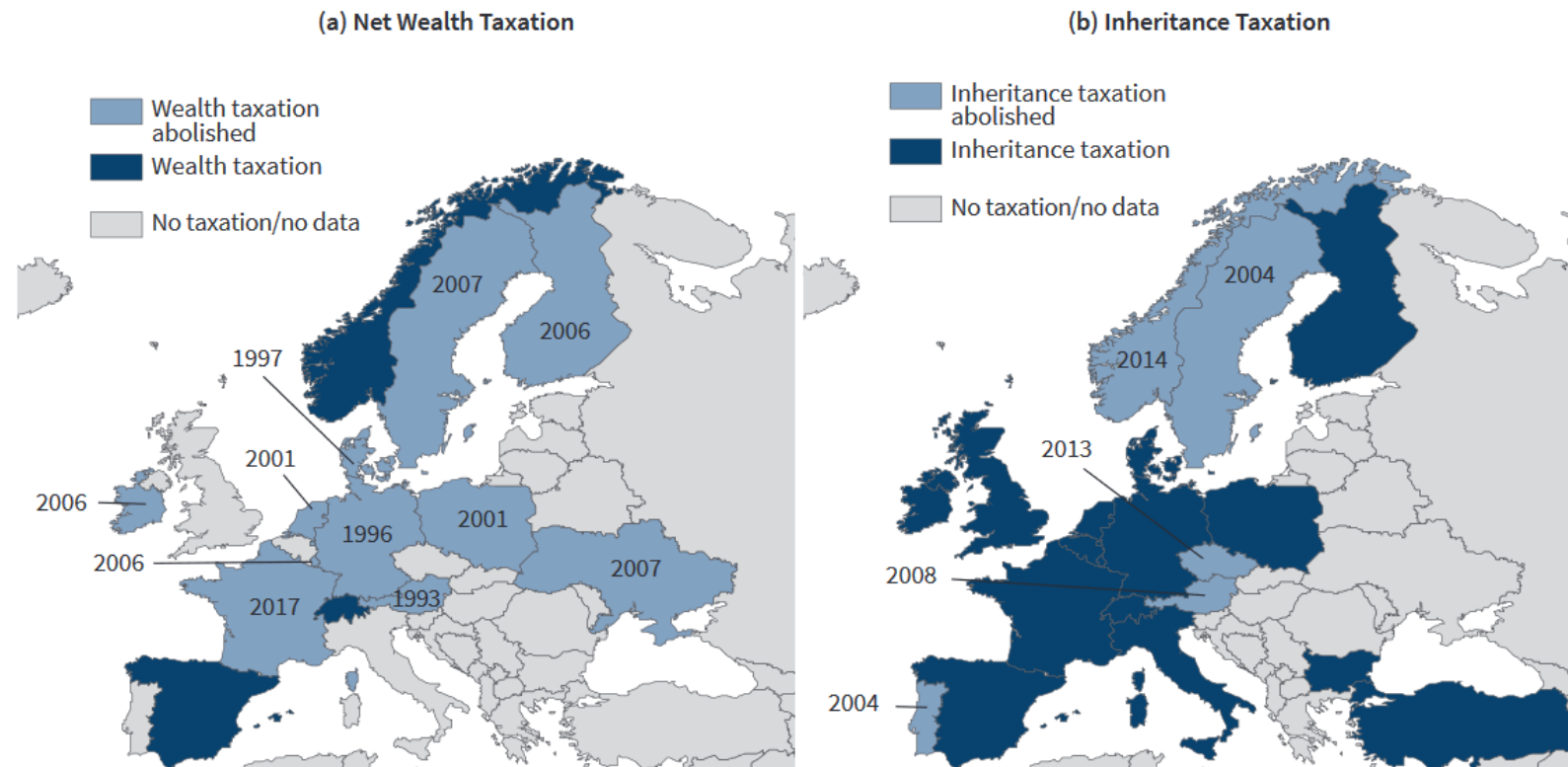


Nevertheless, no Laffer effect → Tax revenues decrease

Wealth Taxation in Europe

Figure 2

Net Wealth and Inheritance Taxation in Europe, 2017/2018



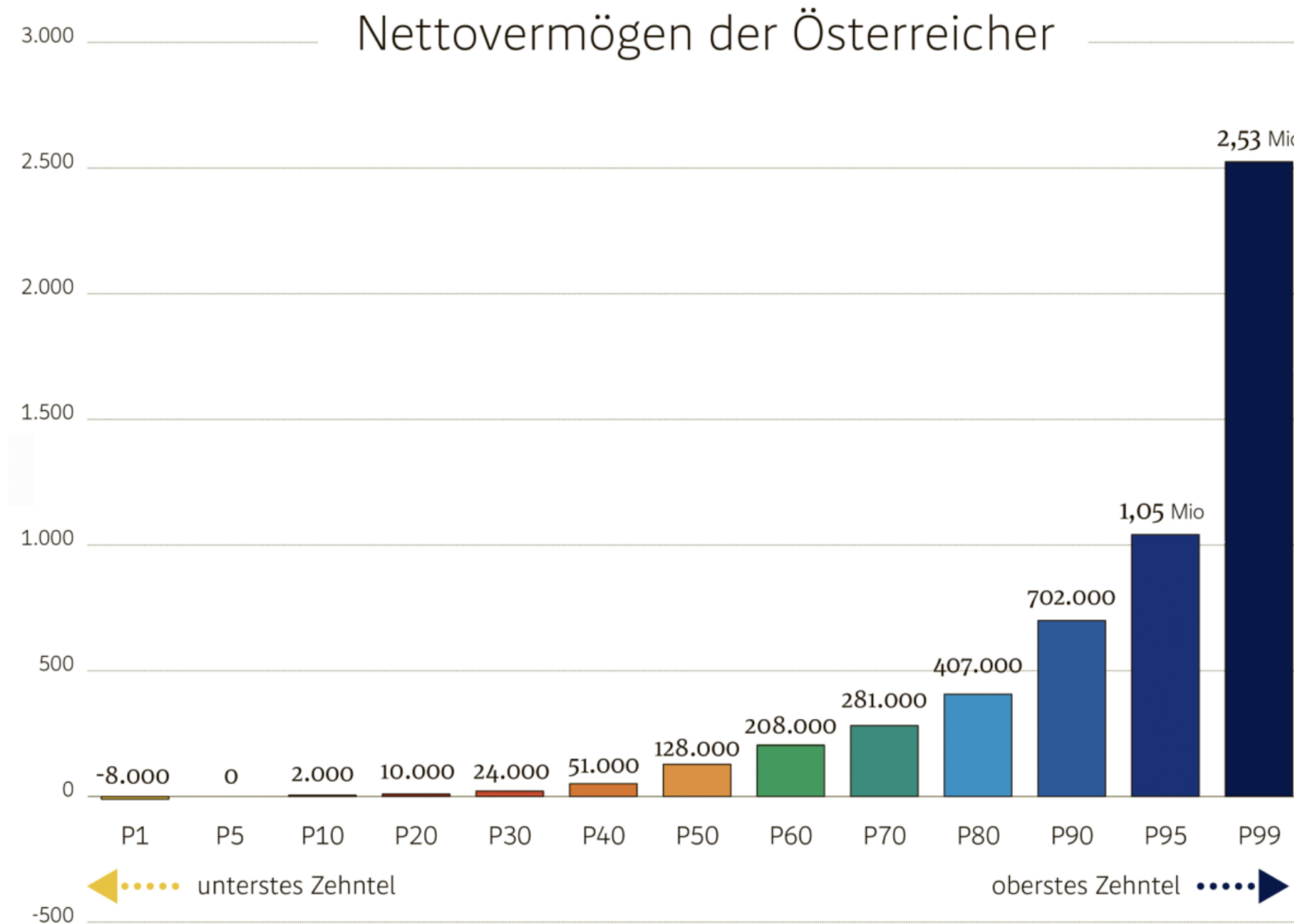
Note: The year marks the date of abolition of the tax.
Source: Authors' compilation of various sources (2018).

© ifo Institute

- 1990: 12 OECD countries with a Wealth Tax.
- 2017/18: Only 3
- Austria: No wealth tax (since 1993) or inheritance tax (since 2008)
- Highly differing tax rates and designs

net wealth

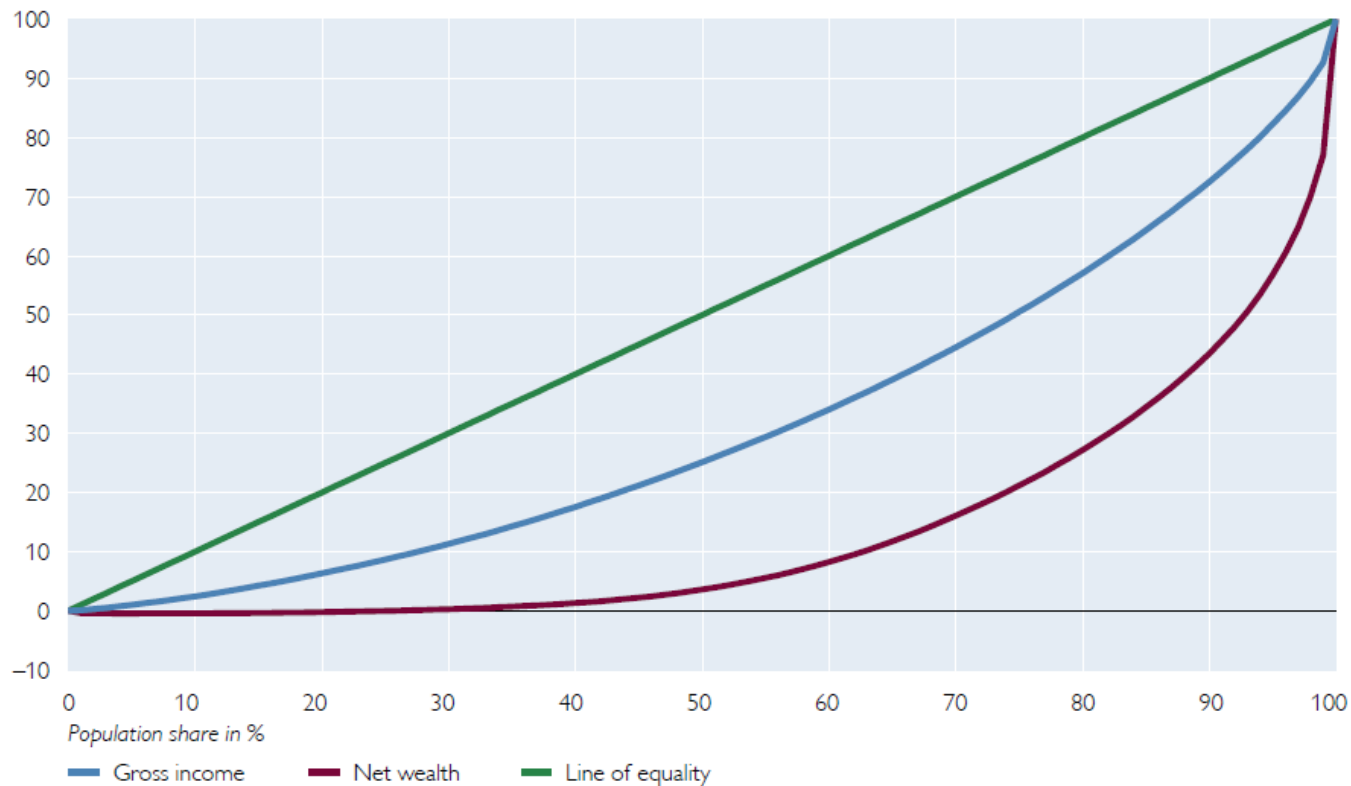
$$= \text{Asset} - \text{liabilities}$$



Inequality in Austria I

Lorenz curves of gross income and net wealth

Gross income and net wealth shares in %



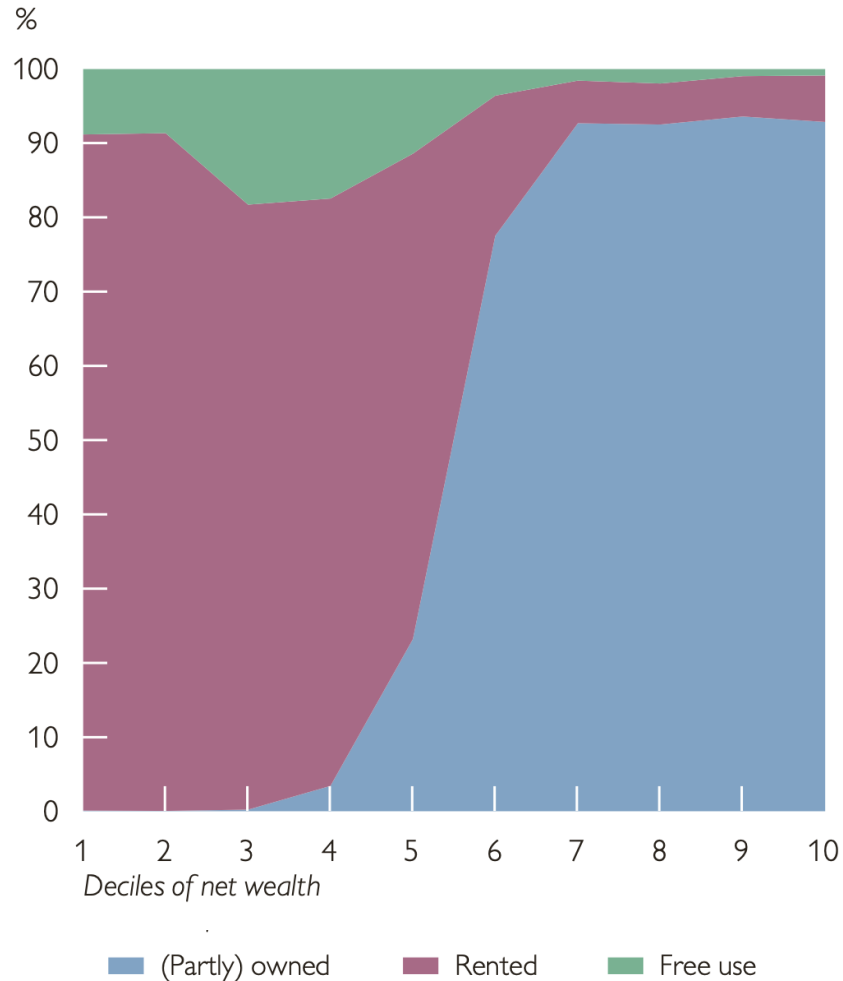
Source: HFCS Austria 2017, OeNB.

- Top 10% own just under 30% of total gross income
- Top 10% own over 50% of total wealth
- Bottom 50% together own under 5% of total wealth

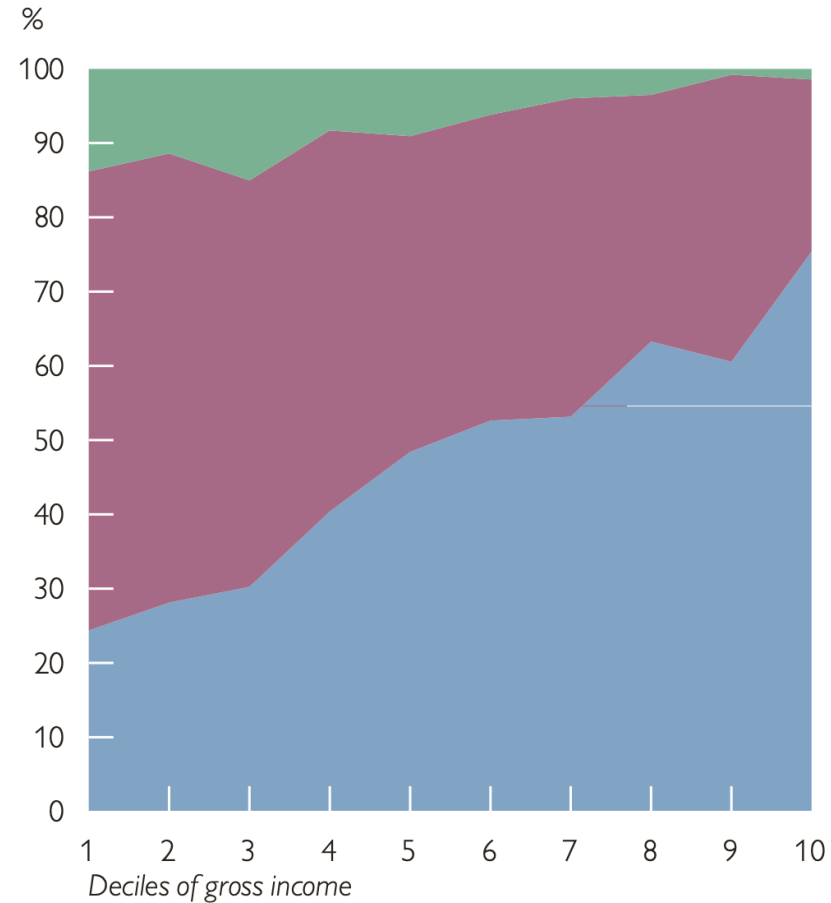
Distribution of Ownership/Rent

Housing status

Housing status across net wealth deciles

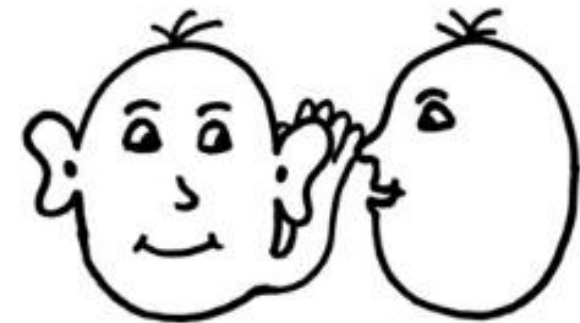


Housing status across gross income deciles



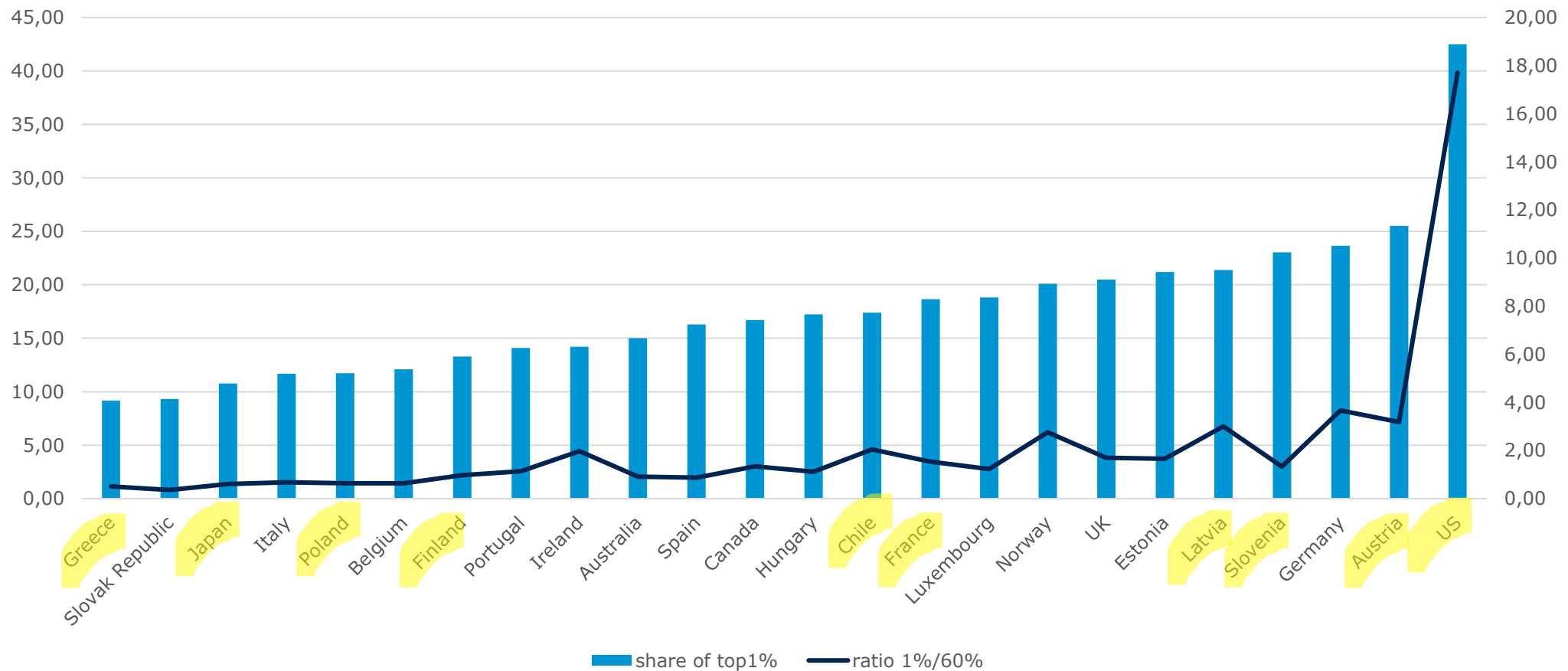
Question

- **Sort the following OECD countries by their wealth inequality as measured by the share of the Top 1% in total wealth:**
- Austria
- Chile
- Finland
- France
- Greece
- Japan
- Slovenia
- US
- Latvia
- Poland



Wealth inequality OECD

Wealth inequality in the OECD



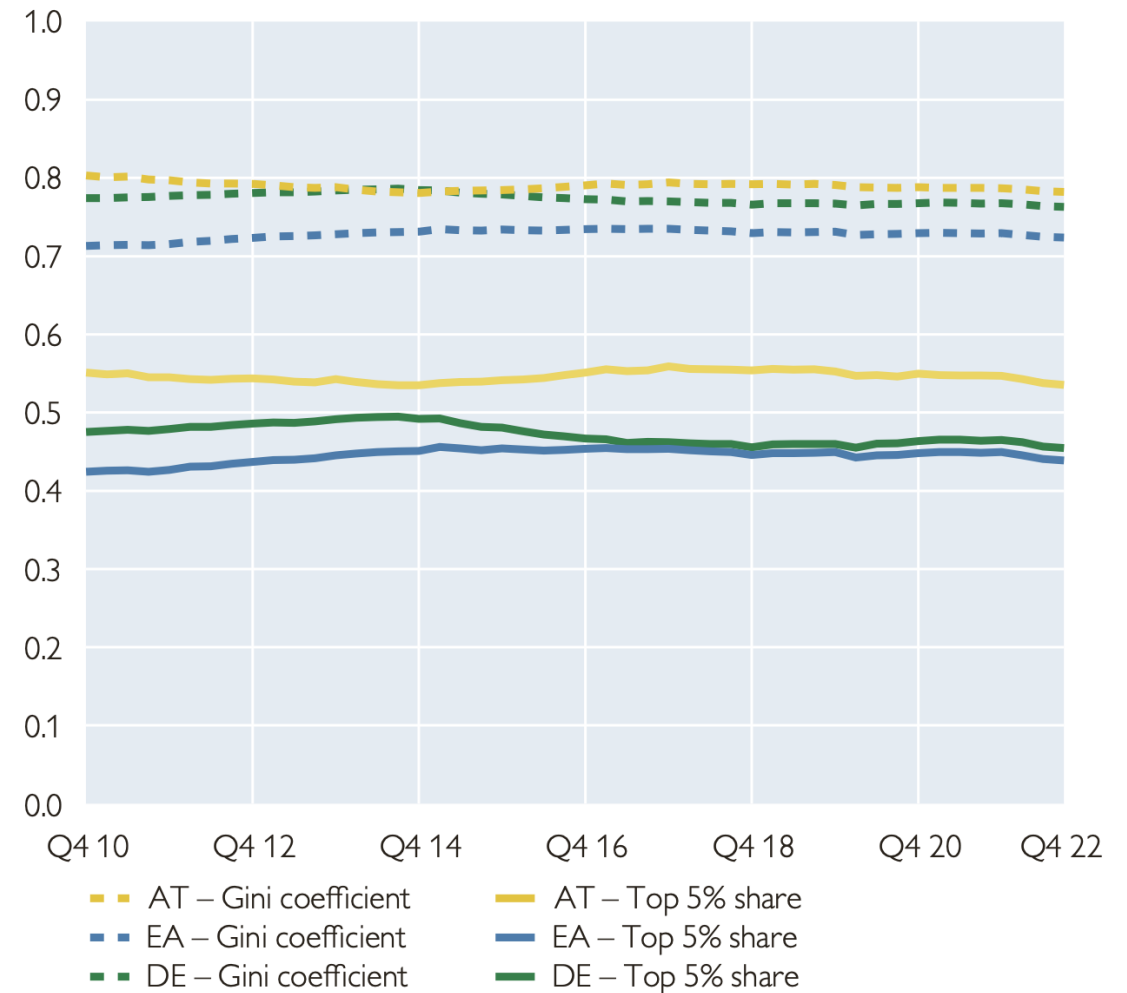
Balestra, C. and R. Tonkin (2018), "Inequalities in household wealth across OECD countries: Evidence from the OECD Wealth Distribution Database", *OECD Statistics Working Papers*, No. 2018/01, OECD Publishing, Paris, <https://doi.org/10.1787/7e1bf673-en>.

Gini Coefficient for Wealth Inequality

- High top 5% share in Austria
- Top 1% in Austria 40% vs. 28% in the EU average
- Gini in Austria is also higher in comparison

Inequality indicators

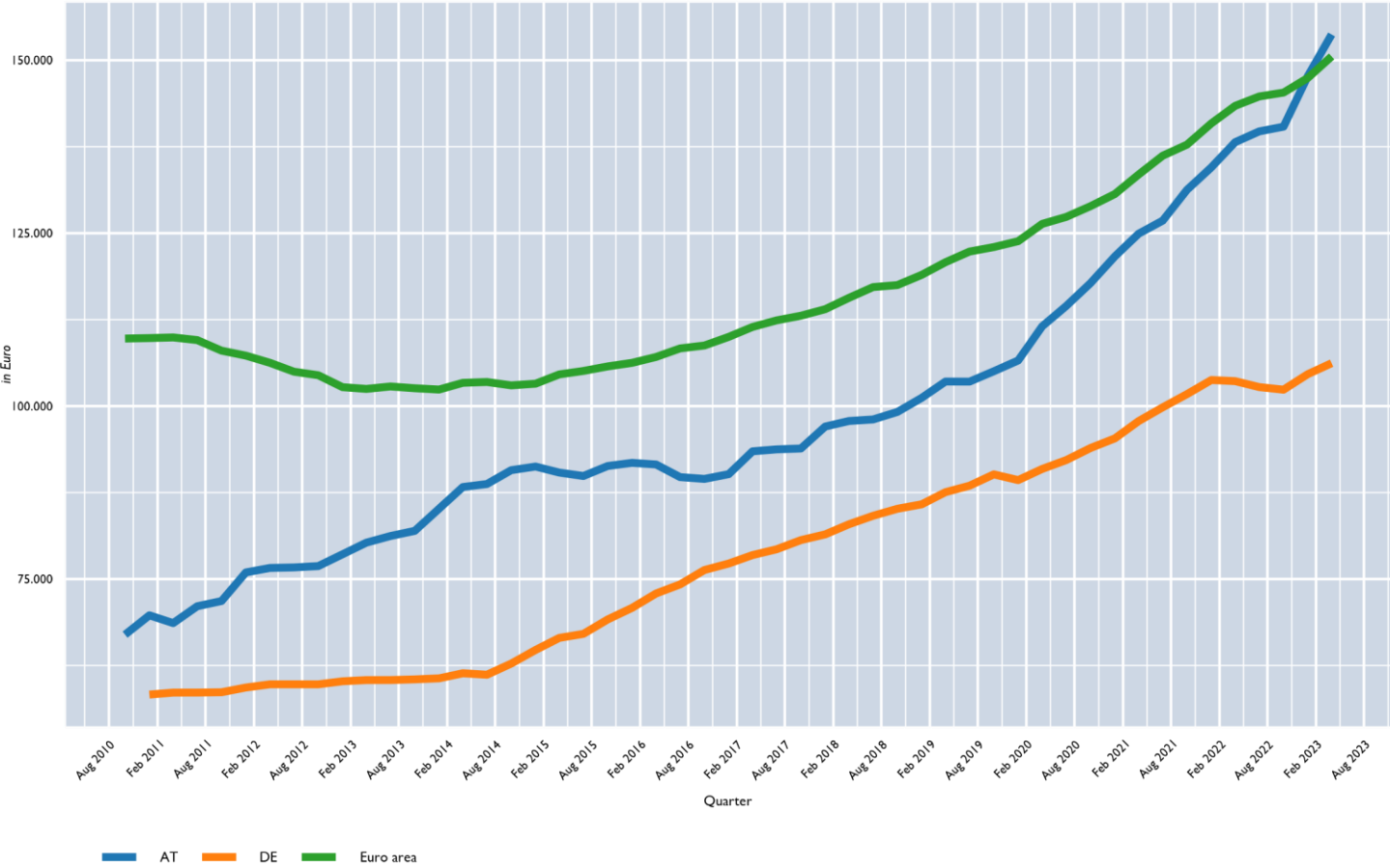
Share; Gini coefficient



Median-Net-wealth

International comparison - Median net wealth

Quarterly data

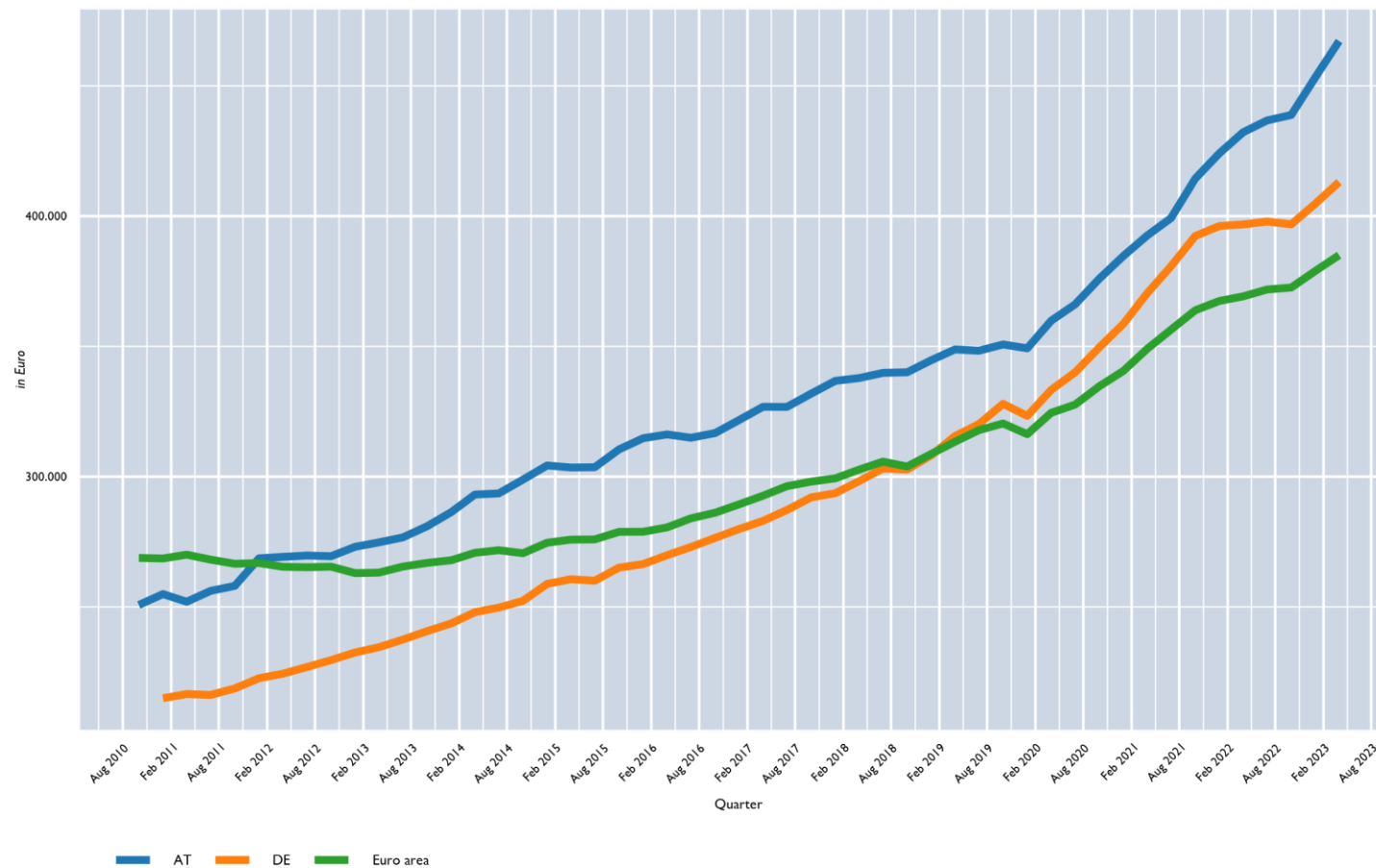


Source: DWA, ECB, OeNB, Jan. 2024

Average-Netto-Wealth

International comparison - Mean net wealth

Quarterly data



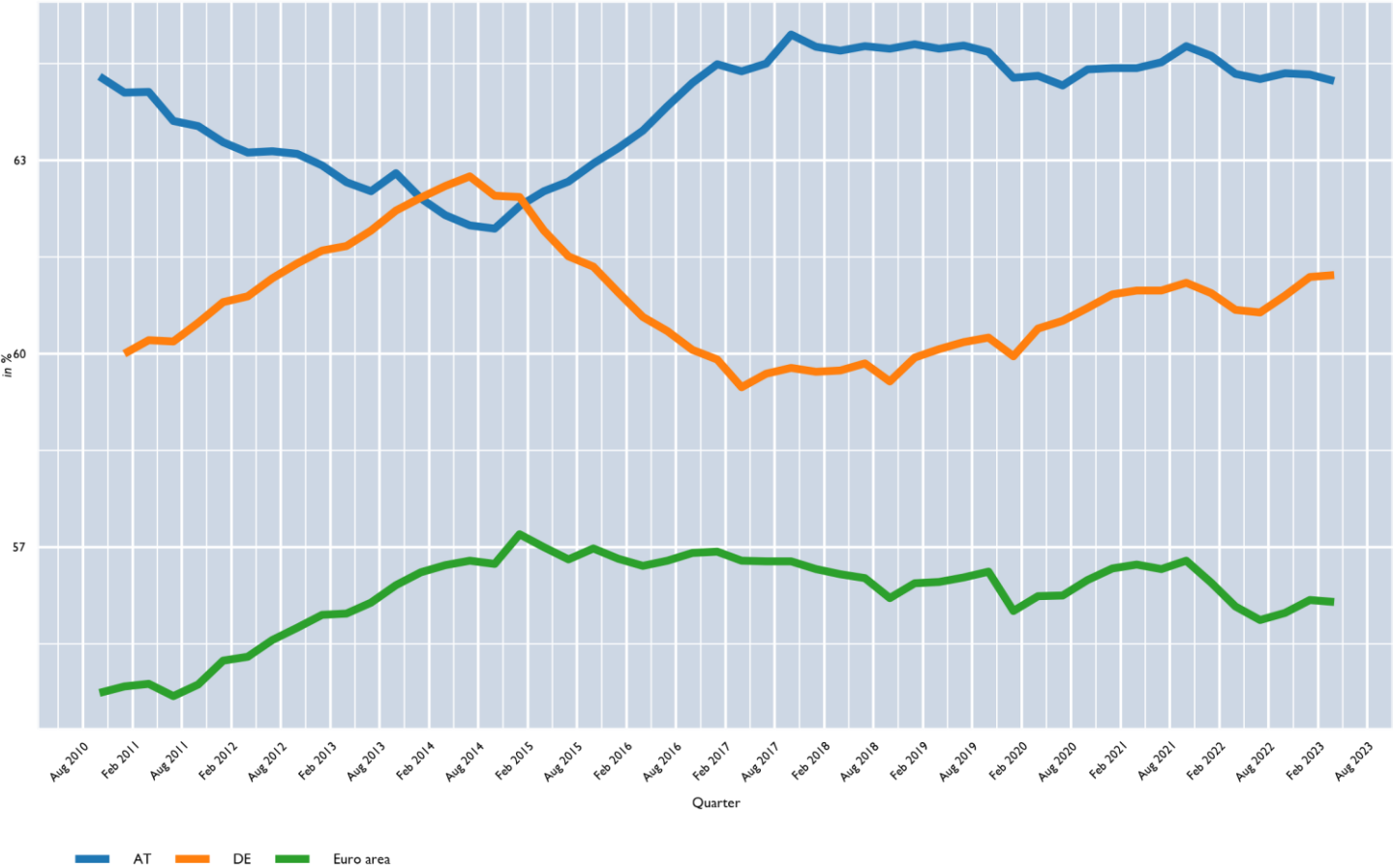
Average net wealth is more extreme than the median

→ Extreme incomes are higher!

Top 10%

International comparison - Top 10%

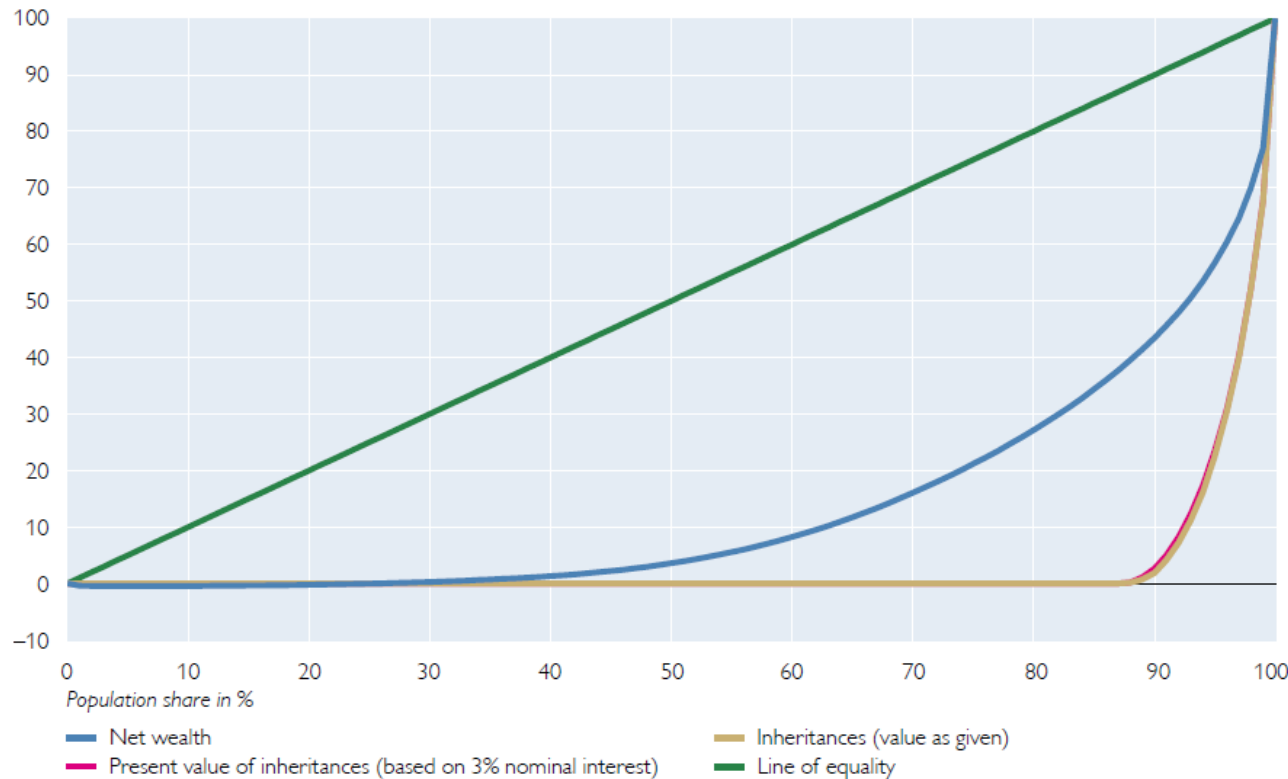
Quarterly data



Inequality in Austria III

Lorenz curves of inheritances and net wealth

Inheritances and net wealth shares in %



Source: HFCS Austria 2017, OeNB.

Inheritances are distributed much more unequally → almost the entire volume is held by the Top 10%

Leaving legacies and liabilities: The distribution of wealth at death

Franziska Disslbacher Severin Rapp (2024, Working Paper)

- What does the current wealth distribution in Austria look like?
- Which individual and institutional factors shape this distribution?

Leaving legacies and liabilities: The distribution of wealth at death

Franziska Disslbacher Severin Rapp (2024, Working Paper)

- What does the current wealth distribution at death look like?
- Gini indices above 1.0!
- Approx. 30% of the deceased leave behind debts (liabilities)
- Which individual and institutional factors shape this distribution?
- Life cycle effects only explain the inequality to a limited extent!
- Apartment/House ownership → high wealth → Intergenerational Inequality

- **Asset situation of the deceased is recorded**, creating data on the assets present at the time of death. (**Legally mandated!!**)
- When **owners of real estate or businesses die**, or the **balance of assets and liabilities (including costs like the burial) is above the *de minimis* limit of 5,000 Euro**, the resulting data is particularly detailed.
- **Digitized: Significant effort!**
- **Data from 2014–2019.**
- **Innovative asset evidence:**
 - Including **valuables and vehicles**, which are often missing in inheritance or tax data!
 - Originally substantial uncertainty regarding asset distribution (often relying on surveys)

Distribution of the Forms of Succession

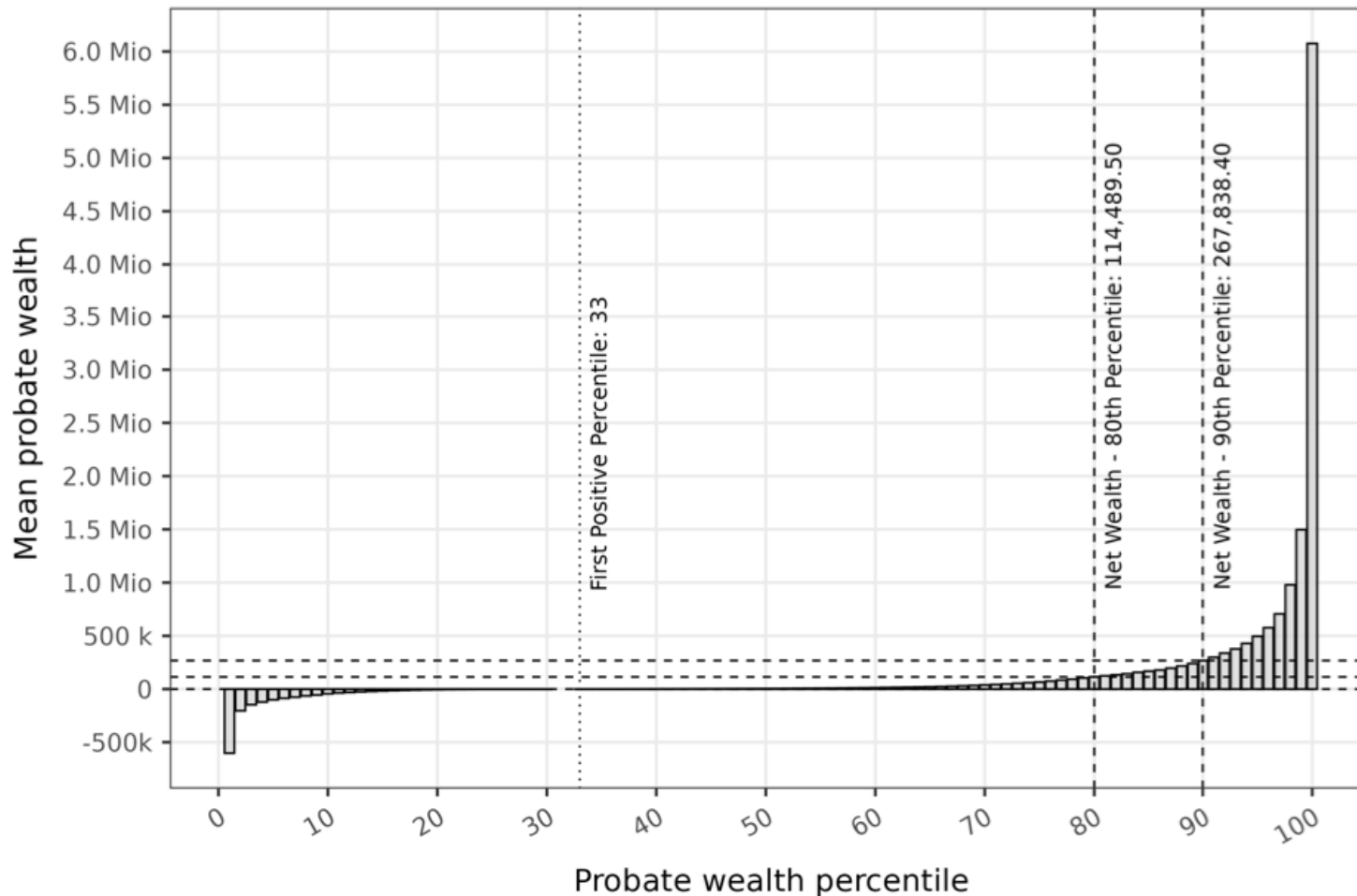
Tabelle 7.1: Anzahl Erbantrittserklärungen gesamt

Erbantrittserklärung	Anzahl	Anteil
An Zahlung statt	21.381	36,97
Unbedingt	19.075	32,99
Bedingt	14.623	25,29
Entschlagung	2.232	3,86
Sonstiges	420	0,73
Legat	96	0,17
Summe	57.827	100,00

Anmerkung: Die Tabelle zeigt die Anzahl der Erbantrittserklärungen von Erb:innen und die Anteile aller Erbantrittserklärungen relativ zur Gesamtsumme. Die Daten stammen aus der Dokumentation von Verlassenschaftsakten im Zeitraum 2014-2019. Quelle: Eigene Berechnung unter Verwendung von Populationsgewichten.

Results

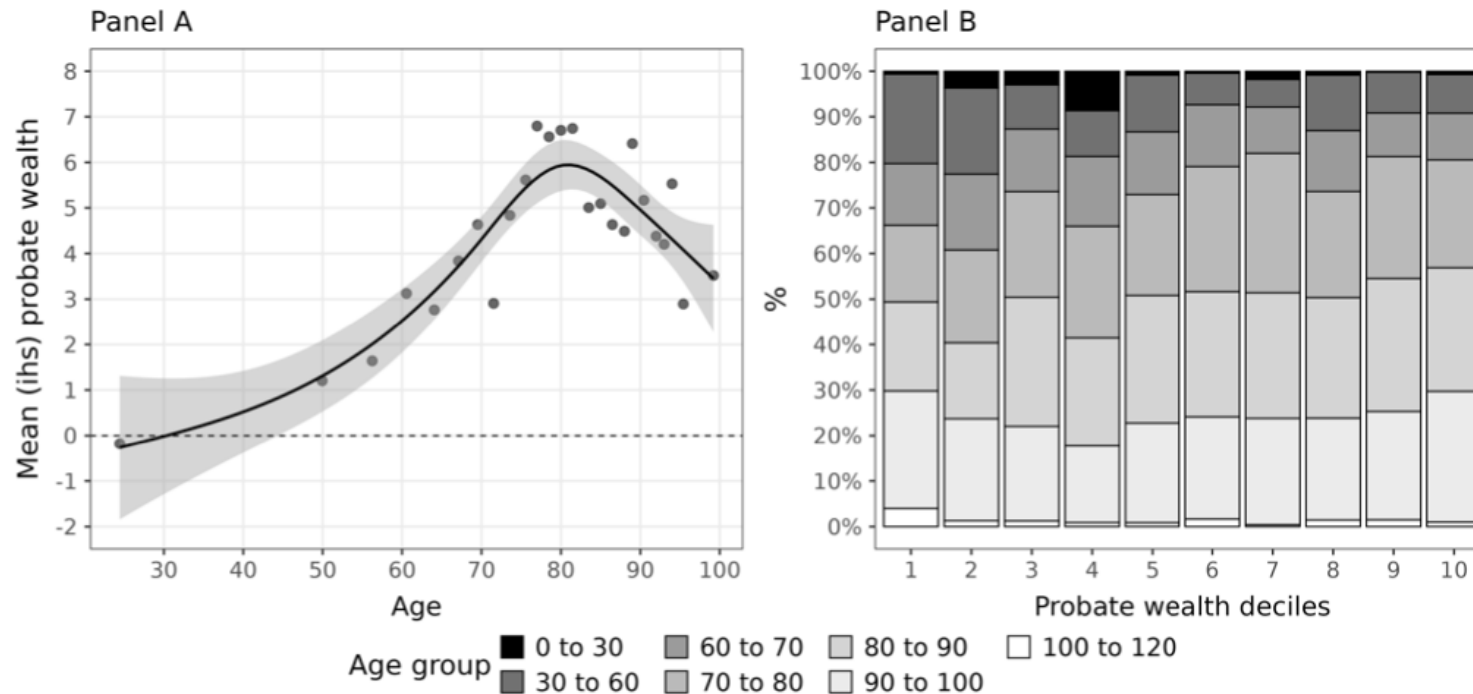
Figure 1: Pen's parade of probate wealth



- Pen's parade of probate wealth
- The richest 0.1% leave behind approx. EUR 22,600,000 !!!
- 30% leave behind negative estates!
- Debts with pharmacies, landlords/landladies, and debt collection agencies are essential drivers.
- Over 90% of the estate wealth is accounted for by the top 10% !

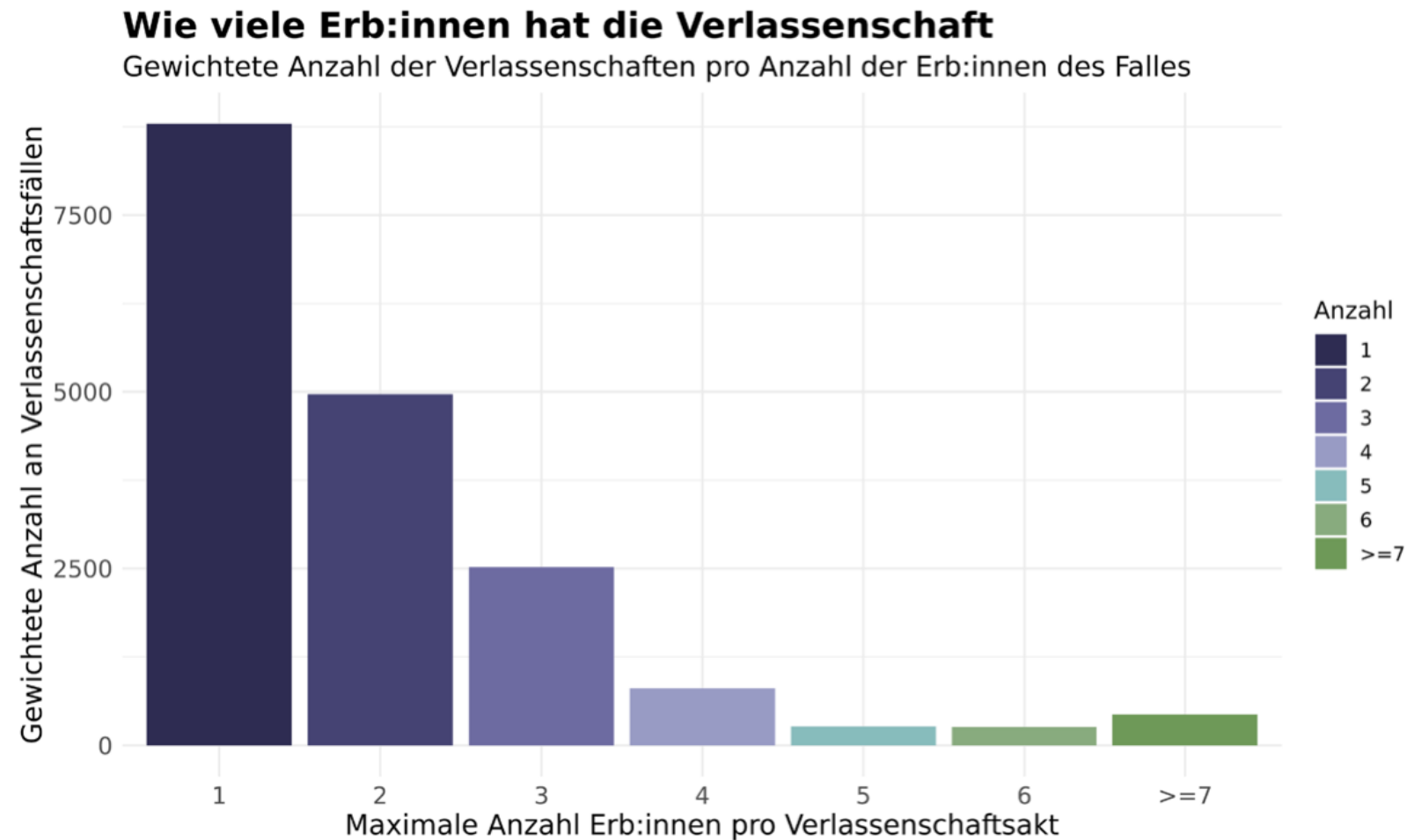
Results

Figure 2: The joint distribution of age and wealth



- Before age 60 and after age 90 $\Rightarrow$ lower wealth!
- Nursing home residents are more frequently over-indebted!
- The *Pflegeregress* (the abolition of the policy in force until 2017 that allowed the state to reclaim long-term care expenses from estates) reduces inequality.

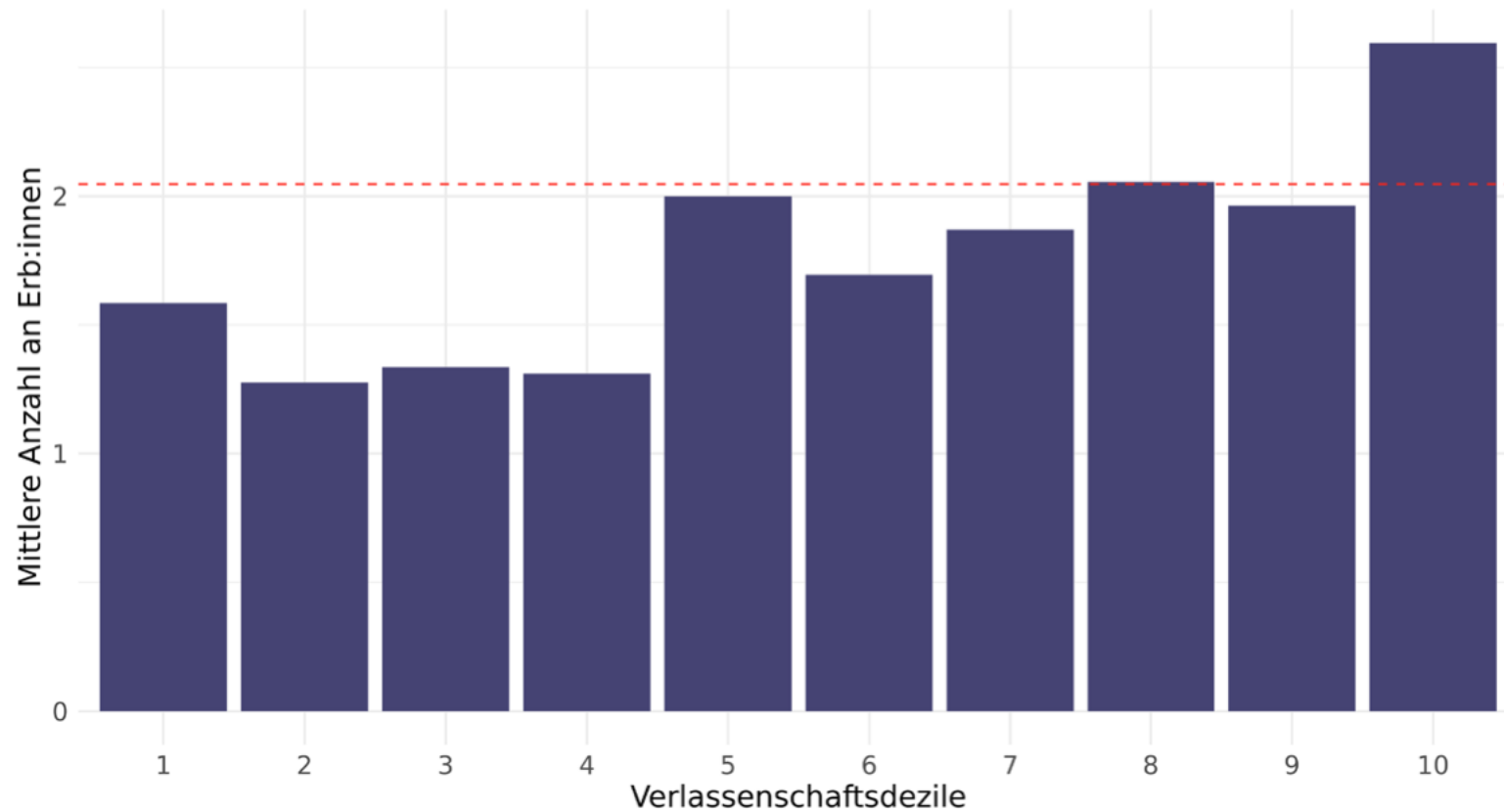
^a Note: The Figure shows a smoothed estimated of mean probate wealth along the age distribution in the first panel (based on a generalized additive model with a cubic spline). Probate wealth is ihs-transformed. The second panel shows for each decile of the probate wealth distribution the share of decedents in a given age group.



Heirs per decile

Wie viele Erb:innen haben Verlassenschaften pro Vermögensdezil

Mittlere Anzahl der Erb:innen pro Vermögensdezil

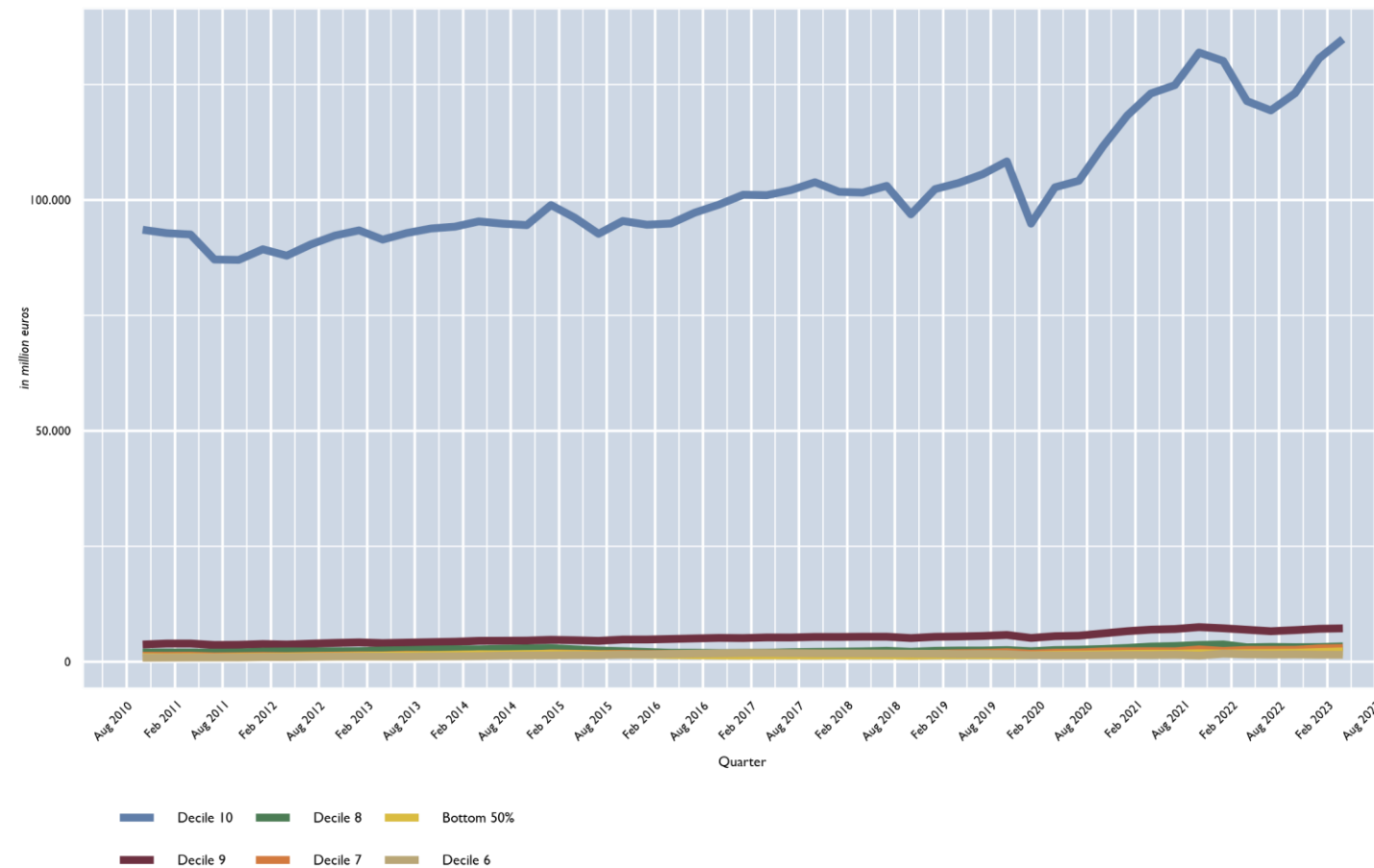


- **Coupling with existing administrative procedures:** "By linking the **inheritance tax** to already existing procedures, such as the **probate process** (Verlassenschaftsverfahren), administrative costs can be minimized."

Financial Investments

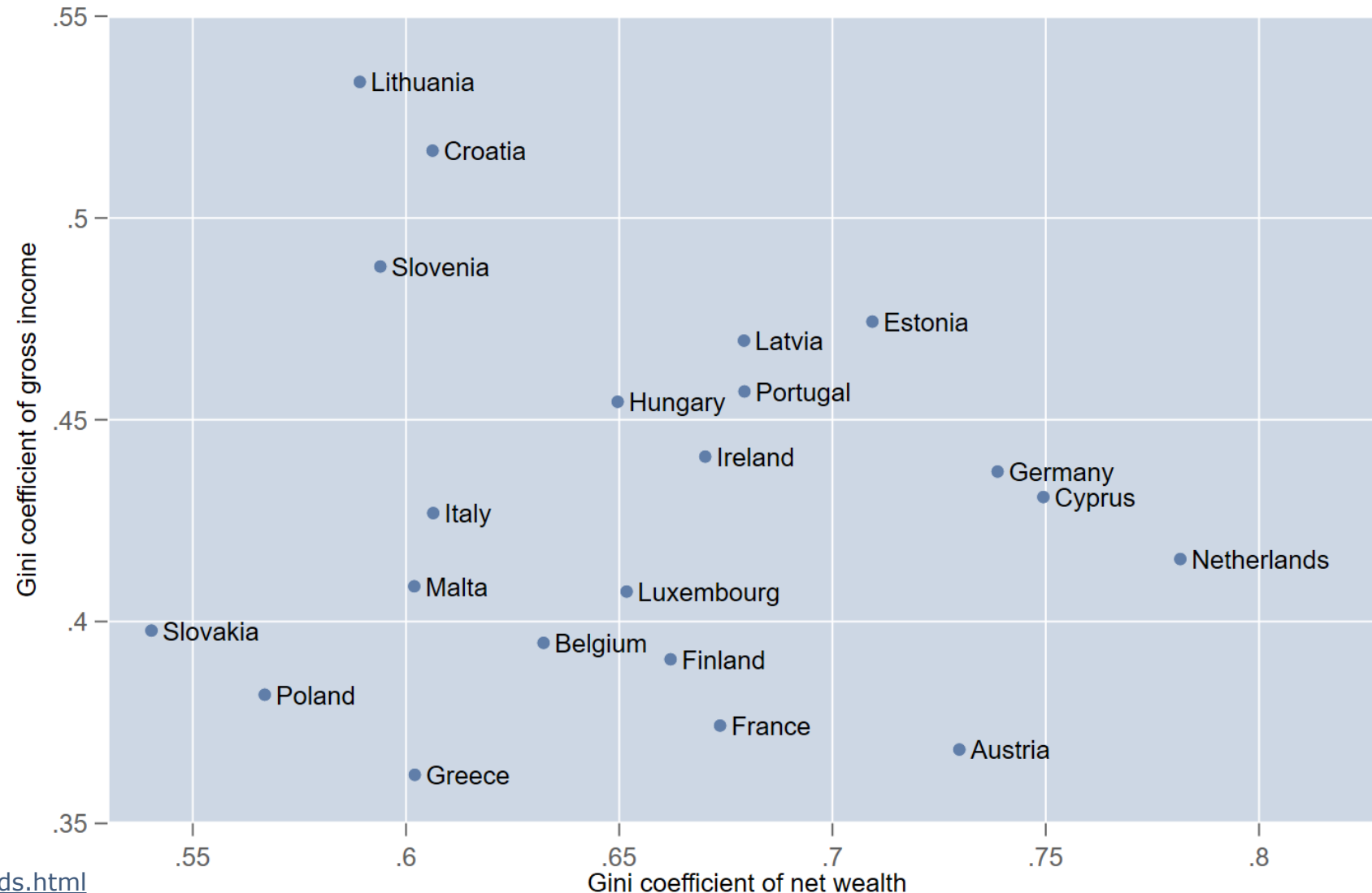
Financial investments in Austria across decile net wealth

Quarterly data



Income and Wealth Inequality EU

- In Austria, income is relatively equally distributed (y-axis), but wealth is very unequally distributed.
- Overall, less correlation than one would expect.



Conclusio

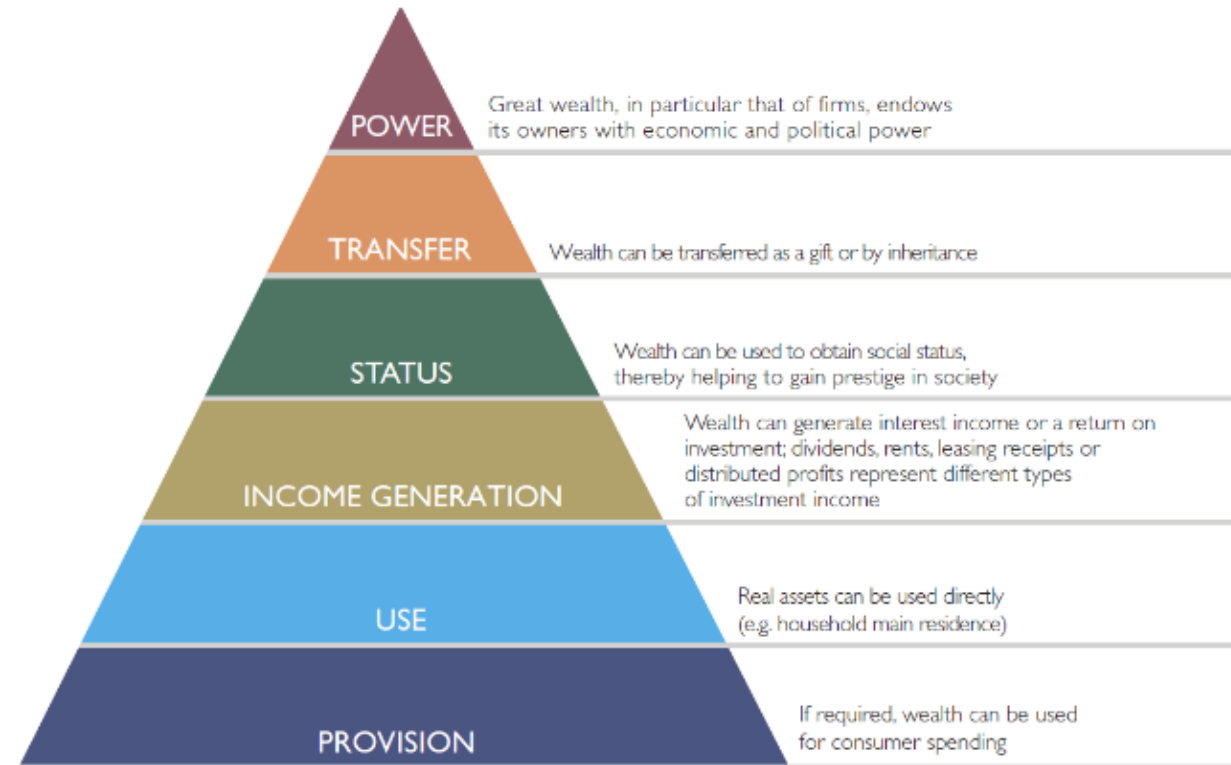
Implications and arguments depend heavily on design and the assets under consideration)

- Private vs. Corporate
- Exemptions
- Progressivity
- Connection to the rest of the tax system
- International cooperation in tax avoidance

And on the objective of the tax!

- **Efficiency gain, Redistribution, Justice, Growth, Tax revenues**

Figure 1: Functions of wealth



Source: Pirmin Fessler & Martin Schürz, 2018. "The functions of wealth: renters, owners and capitalists across Europe and the United States," Working Papers 223, Oesterreichische Nationalbank (Austrian Central Bank)

- Paper: *You´ve Got Mail: A Randomized Field Experiment on Tax Evasion* (Bott et al., 2020)
 - *Was ist ein RCT? Vor- und Nachteile.*
 - *Kapital*



Weiterführende Links

- OECD BEPS: <https://www.oecd.org/tax/beps/beps-actions/>
- BEPS Monitoring Group: <https://bepsmonitoringgroup.wordpress.com/2015/10/05/overall-evaluation/>
- EU Blacklist Press Release: [https://europa.eu/rapid/press-release MEMO-19-1629 en.htm](https://europa.eu/rapid/press-release_MEMO-19-1629_en.htm)
- EU Blacklist: [https://ec.europa.eu/taxation customs/tax-common-eu-list en](https://ec.europa.eu/taxation_customs/tax-common-eu-list_en)
- Oxfam über EU Blacklist: https://www-cdn.oxfam.org/s3fs-public/file_attachments/bn-blacklist-whitewash-tax-havens-eu-281117-en_0.pdf
- Tax Justice Network über EU Blacklist: <https://www.taxjustice.net/2017/11/27/blacklisting-the-eu-paradise-lost/>
- Tax Justice Network: <https://www.taxjustice.net/>
- Interview mit Zucman (Autor des Papers): <https://www.bloomberg.com/news/articles/2019-10-15/brain-behind-wealth-tax-makes-new-case-for-soaking-billionaires>
- Indizes für Steueroasen: <https://www.corporatetaxhavenindex.org/>,
<https://www.financialsecrecyindex.com/en/>
- LSE Blog: <https://blogs.lse.ac.uk/eurompblog/2015/05/18/europes-hidden-economy-how-governments-can-bring-undeclared-work-out-of-the-shadows/>